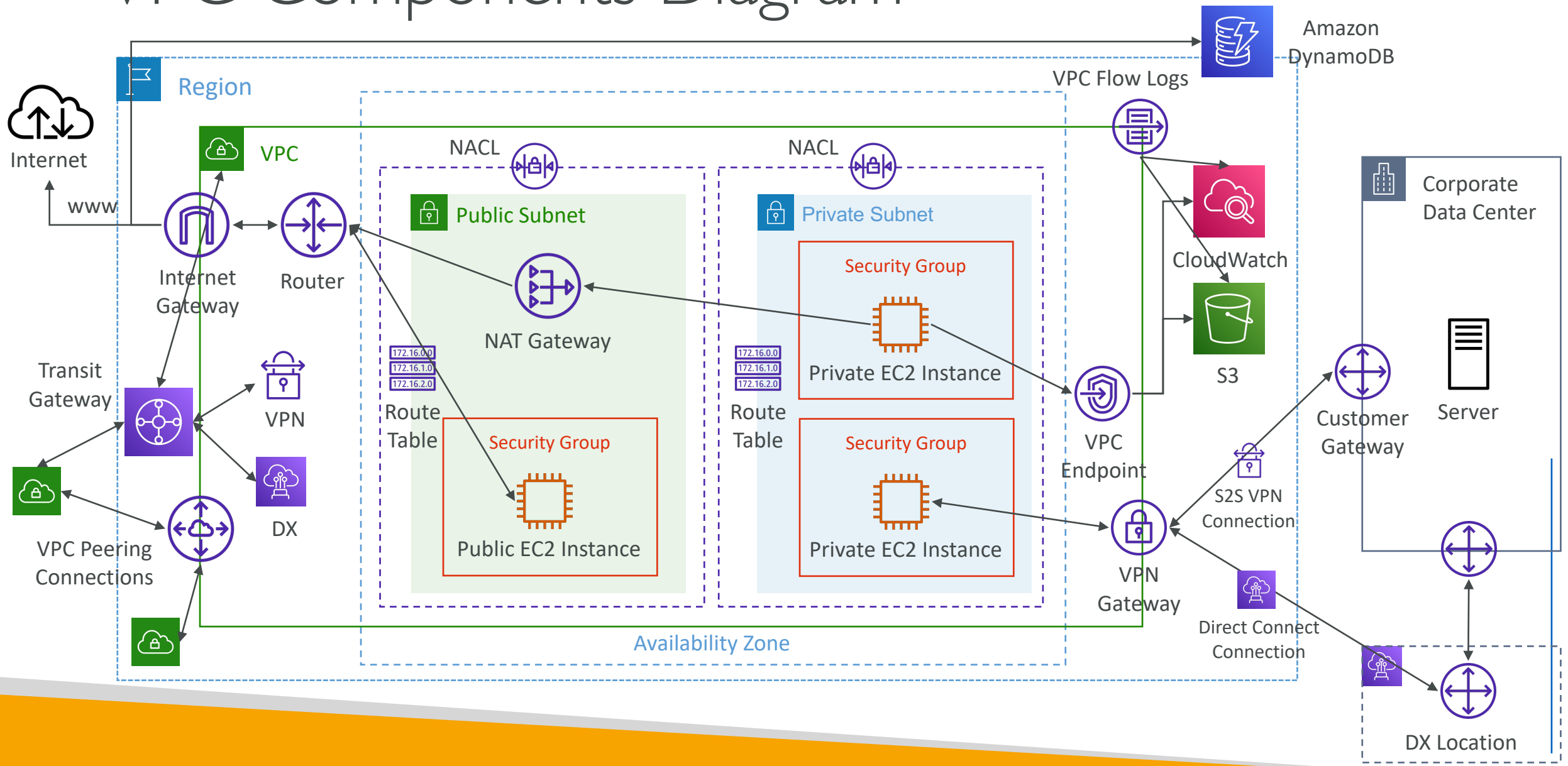
Amazon VPC

VPC Components Diagram



Understanding CIDR – IPv4

- **Classless Inter-Domain Routing** – a method for allocating IP addresses
- Used in **Security Groups** rules and AWS networking in general

IP version	Type	Protocol	Port range	Source	Description
IPv4	SSH	TCP	22	122.149.196.85/32	–
IPv4	HTTP	TCP	80	0.0.0.0/0	–

- They help to define an IP address range:
  - We've seen WW.XX.YY.ZZ**/32** => one IP
  - We've seen 0.0.0.0**/0** => all IPs
  - But we can define: 192.168.0.0**/26** =>192.168.0.0 – 192.168.0.63 (64 IP addresses)

Understanding CIDR – IPv4

- A CIDR consists of two components
- **Base IP**
  - Represents an IP contained in the range (XX.XX.XX.XX)
  - Example: 10.0.0.0, 192.168.0.0, …
- **Subnet Mask**
  - Defines how many bits can change in the IP
  - Example: /0, /24, /32
  - Can take two forms:
    - /8 ⇔ 255.0.0.0
    - /16 ⇔ 255.255.0.0
    - /24 ⇔ 255.255.255.0
    - /32 ⇔ 255.255.255.255

Understanding CIDR – Subnet Mask

- The Subnet Mask basically allows part of the underlying IP to get additional next values from the base IP

192 . 168 . 0 . 0 **/32 =>** allows for **1** IP (2^0) → 192.168.0.0

192 . 168 . 0 . 0 **/31 =>** allows for **2** IP (2^1) → 192.168.0.0 -> 192.168.0.1

192 . 168 . 0 . 0 **/30 =>** allows for **4** IP (2^2) → 192.168.0.0 -> 192.168.0.3

192 . 168 . 0 . 0 **/29 =>** allows for **8** IP (2^3) → 192.168.0.0 -> 192.168.0.7

192 . 168 . 0 . 0 **/28 =>** allows for **16** IP (2^4) → 192.168.0.0 -> 192.168.0.15

192 . 168 . 0 . 0 **/27 =>** allows for **32** IP (2^5) → 192.168.0.0 -> 192.168.0.31

192 . 168 . 0 . 0 **/26 =>** allows for **64** IP (2^6) → 192.168.0.0 -> 192.168.0.63

192 . 168 . 0 . 0 **/25 =>** allows for **128** IP (2^7) → 192.168.0.0 -> 192.168.0.127

192 . 168 . 0 . 0 **/24 =>** allows for **256** IP (2^8) → 192.168.0.0 -> 192.168.0.255

...

192 . 168 . 0 . 0 **/16 =>** allows for **65,536** IP (2^{16}) → 192.168.0.0 -> 192.168.255.255

...

192 . 168 . 0 . 0 **/0 =>** allows for **All IPs** → 0.0.0.0 -> 255.255.255.255



Quick Memo

Octets

1st . 2nd . 3rd . 4th

- **/32** – no octet can change
- **/24** – last octet can change
- **/16** – last 2 octets can change
- **/8** – last 3 octets can change
- **/0** – all octets can change

Understanding CIDR – Little Exercise

- 192.168.0.0/24 = … ?
  - 192.168.0.0 – 192.168.0.255 (256 IPs)
- 192.168.0.0/16 = … ?
  - 192.168.0.0 – 192.168.255.255 (65,536 IPs)
- 134.56.78.123/32 = … ?
  - Just 134.56.78.123
- 0.0.0.0/0
  - All IPs!

- When in doubt, use this website https://www.ipaddressguide.com/cidr

Public vs. Private IP (IPv4)

- The Internet Assigned Numbers Authority (IANA) established certain blocks of IPv4 addresses for the use of private (LAN) and public (Internet) addresses

- **Private IP** can only allow certain values:
  - 10.0.0.0 – 10.255.255.255 (10.0.0.0/8) ← in big networks
  - 172.16.0.0 – 172.31.255.255 (172.16.0.0/12) ← AWS default VPC in that range
  - 192.168.0.0 – 192.168.255.255 (192.168.0.0/16) ← e.g., home networks

- All the rest of the IP addresses on the Internet are Public

Default VPC Walkthrough

- All new AWS accounts have a default VPC
- New EC2 instances are launched into the default VPC if no subnet is specified
- Default VPC has Internet connectivity and all EC2 instances inside it have public IPv4 addresses
- We also get a public and a private IPv4 DNS names

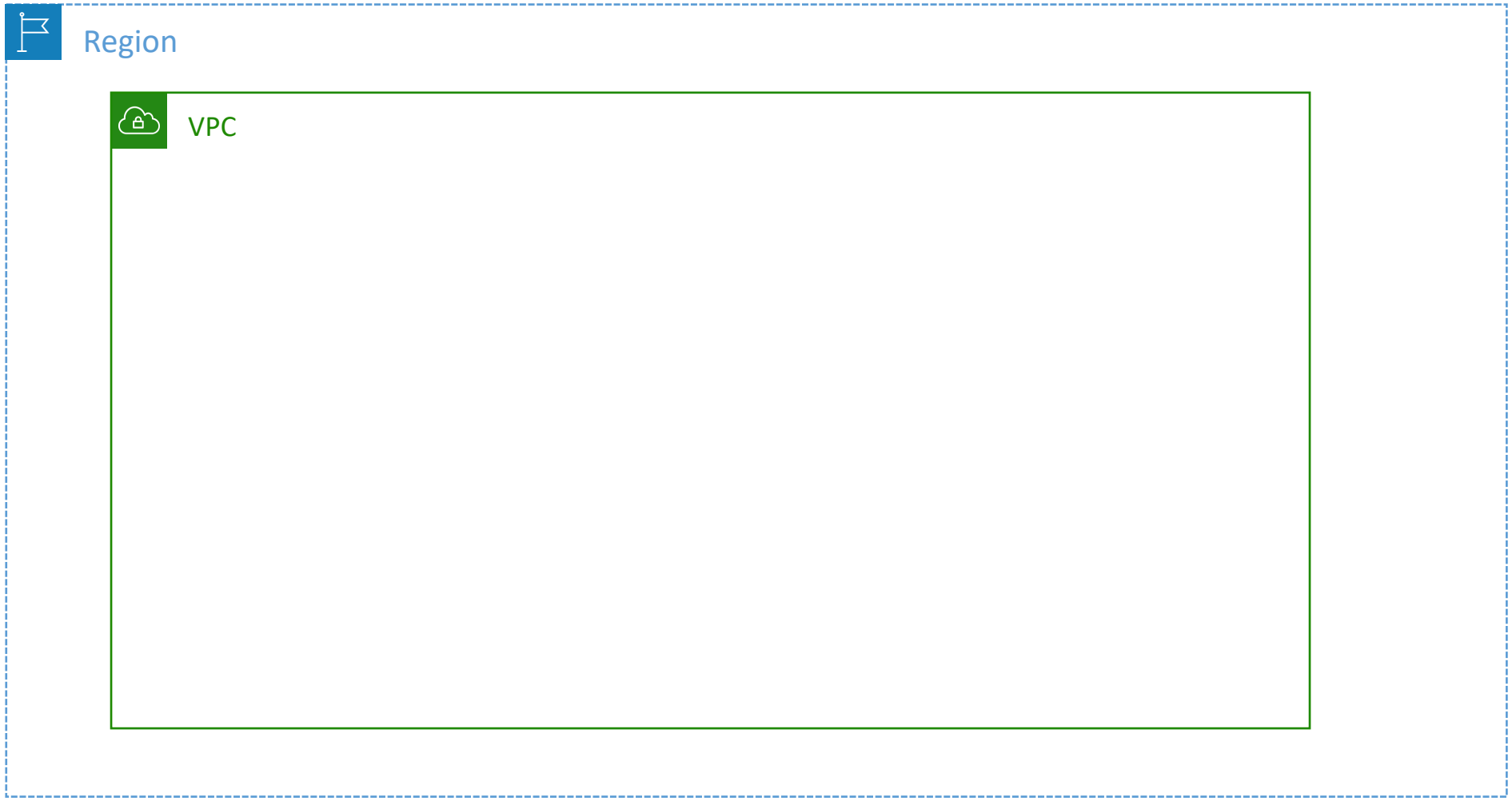
VPC in AWS – IPv4



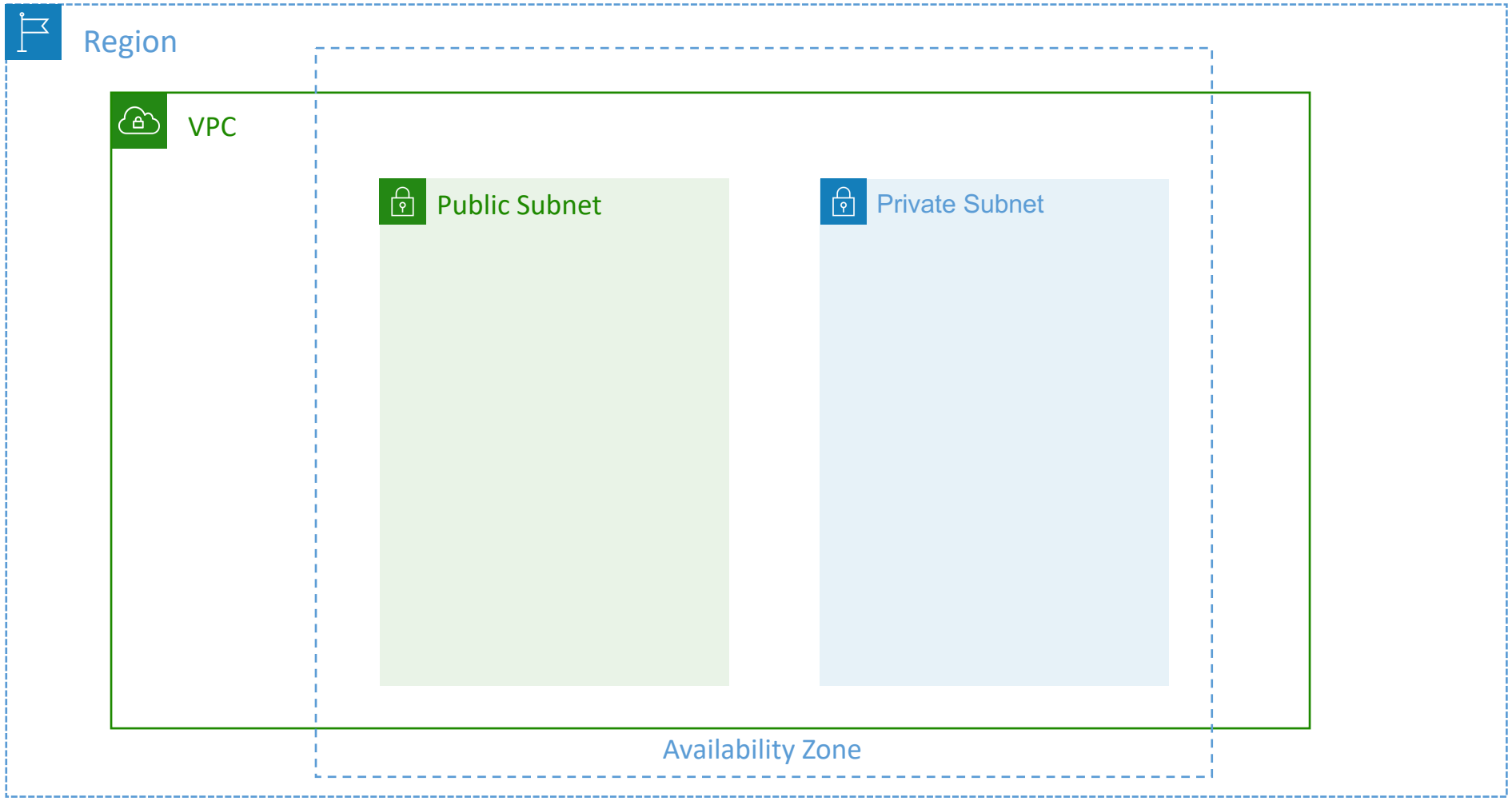
- **VPC = Virtual Private Cloud**
- You can have multiple VPCs in an AWS region (max. 5 per region – soft limit)
- Max. CIDR per VPC is 5, for each CIDR:
  - **Min. size is /28 (16 IP addresses)**
  - **Max. size is /16 (65536 IP addresses)**
- Because VPC is private, only the Private IPv4 ranges are allowed:
  - 10.0.0.0 – 10.255.255.255 (10.0.0.0/8)
  - 172.16.0.0 – 172.31.255.255 (172.16.0.0/12)
  - 192.168.0.0 – 192.168.255.255 (192.168.0.0/16)

- **Your VPC CIDR should NOT overlap with your other networks (e.g., corporate)**

State of Hands-on



Adding Subnets



VPC – Subnet (IPv4)



- AWS reserves **5 IP addresses (first 4 & last 1)** in each subnet
- These 5 IP addresses are not available for use and can't be assigned to an EC2 instance
- Example: if CIDR block 10.0.0.0/24, then reserved IP addresses are:
  - **10.0.0.0** – Network Address
  - **10.0.0.1** – reserved by AWS for the VPC router
  - **10.0.0.2** – reserved by AWS for mapping to Amazon-provided DNS
  - **10.0.0.3** – reserved by AWS for future use
  - **10.0.0.255** – Network Broadcast Address. AWS does not support broadcast in a VPC, therefore the address is reserved

- **Exam Tip,** if you need 29 IP addresses for EC2 instances:
  - You can't choose a subnet of size /27 (32 IP addresses, 32 – 5 = 27 < 29)
  - You need to choose a subnet of size /26 (64 IP addresses, 64 – 5 = 59 > 29)

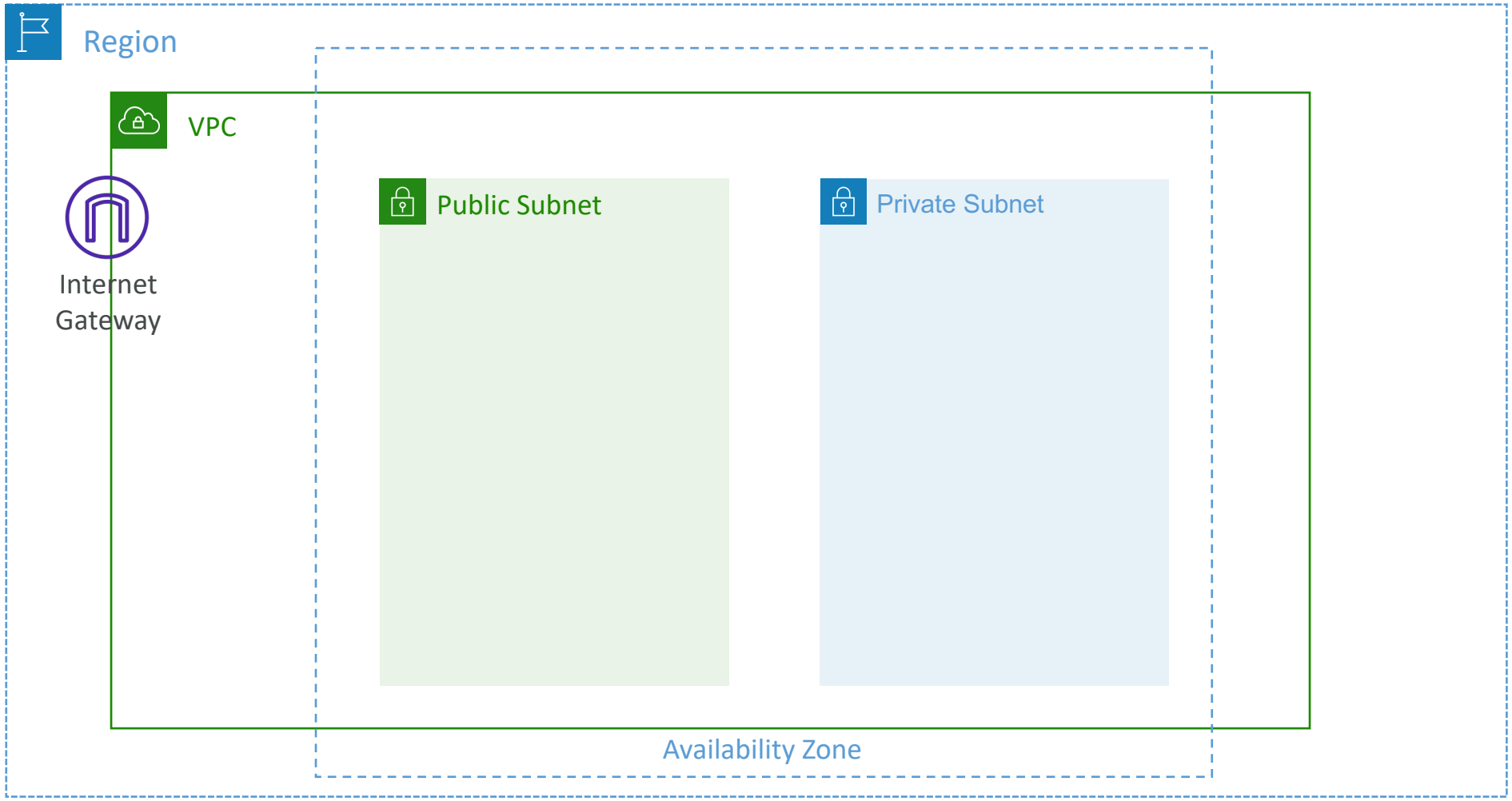
Internet Gateway (IGW)



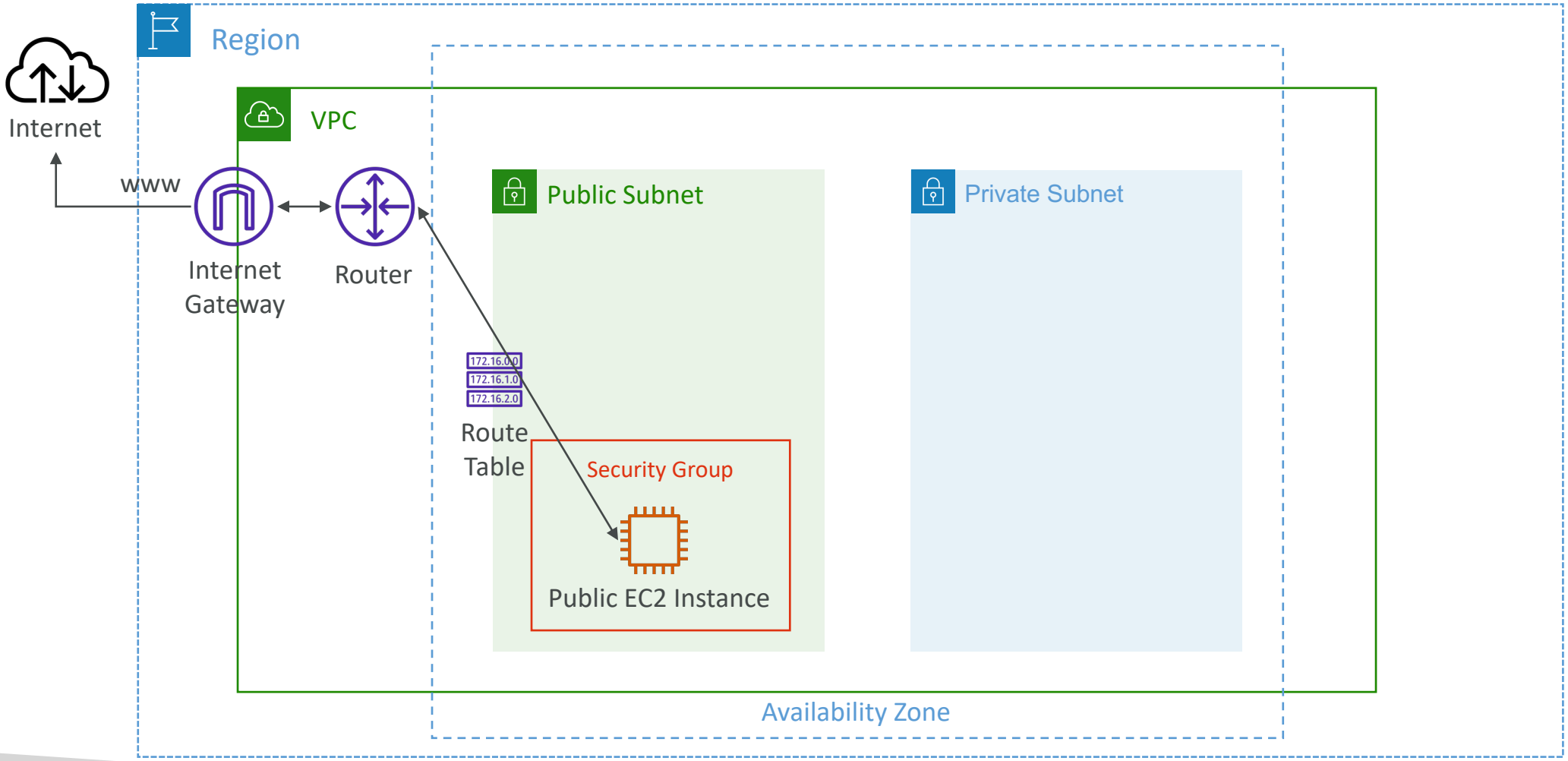
- Allows resources (e.g., EC2 instances) in a VPC connect to the Internet
- It scales horizontally and is highly available and redundant
- Must be created separately from a VPC
- One VPC can only be attached to one IGW and vice versa

- Internet Gateways on their own do not allow Internet access…
- Route tables must also be edited!

Adding Internet Gateway

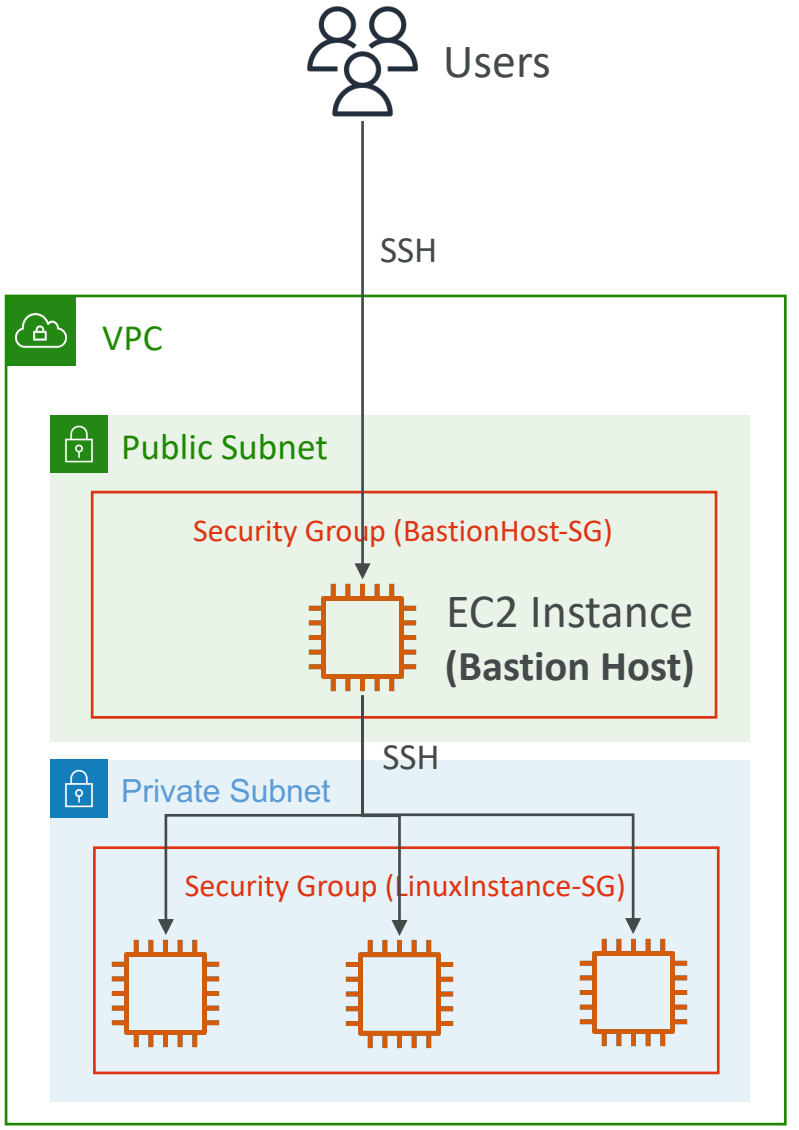


Editing Route Tables



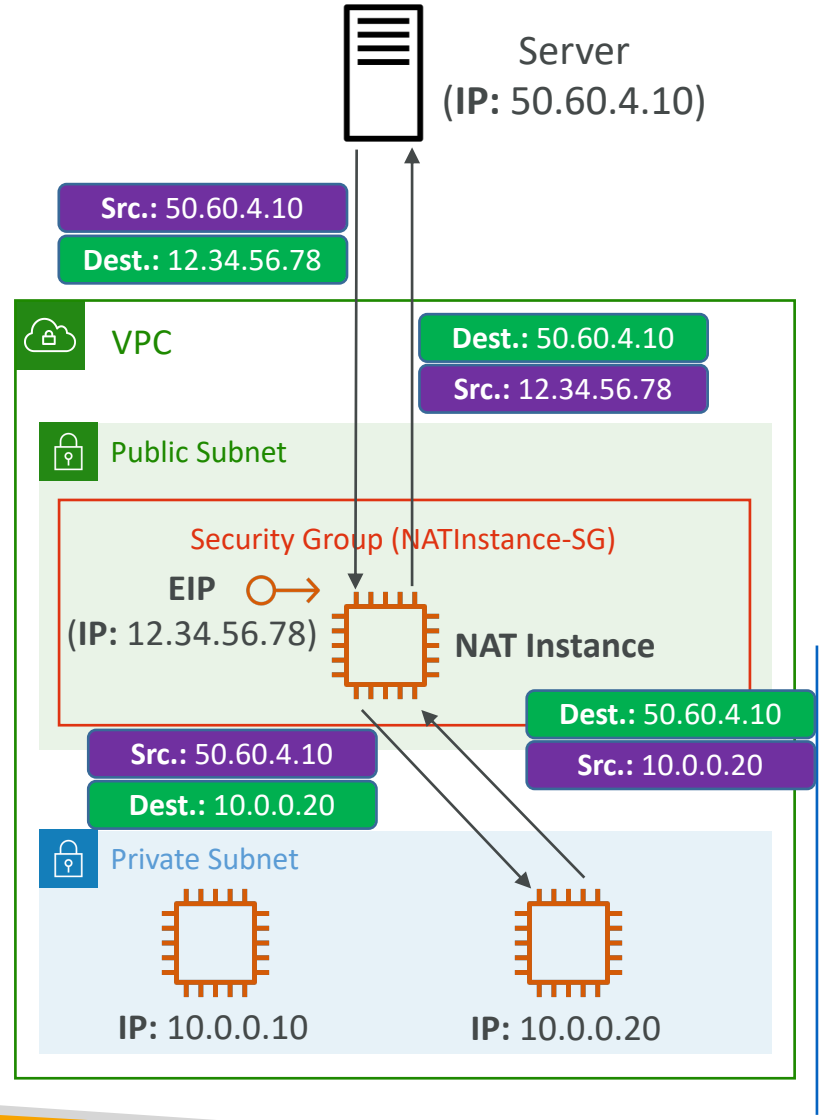
Bastion Hosts

- We can use a Bastion Host to SSH into our private EC2 instances
- The bastion is in the public subnet which is then connected to all other private subnets
- **Bastion Host security group must allow** inbound from the internet on port 22 from restricted CIDR, for example the public CIDR of your corporation
- **Security Group of the EC2 Instances** must allow the Security Group of the Bastion Host, or the private IP of the Bastion host

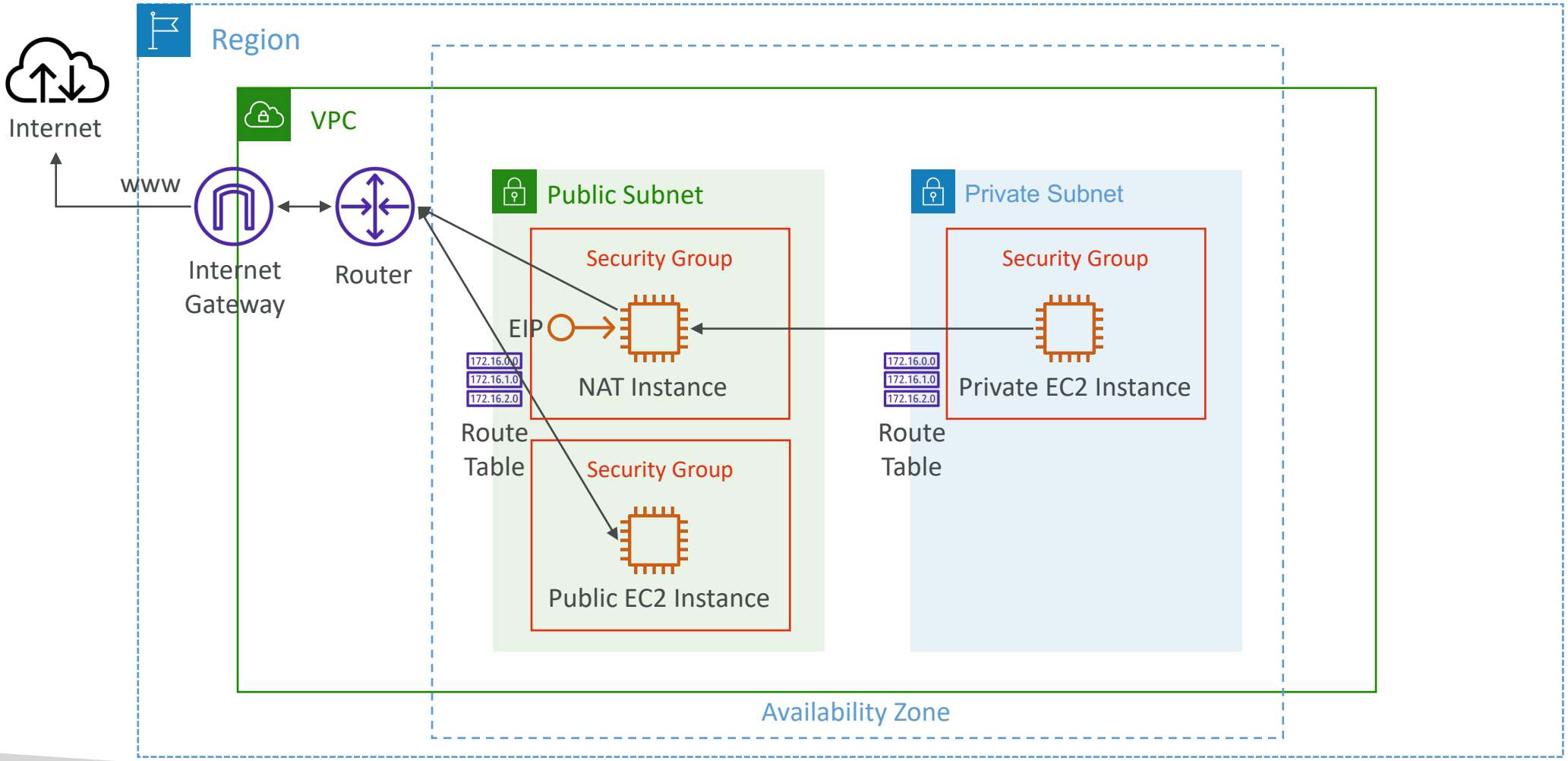


NAT Instance (**outdated**, but still at the exam)

- **NAT = Network Address Translation**
- Allows EC2 instances in private subnets to connect to the Internet
- Must be launched in a public subnet
- Must disable EC2 setting: **Source / destination Check**
- Must have Elastic IP attached to it
- Route Tables must be configured to route traffic from private subnets to the NAT Instance



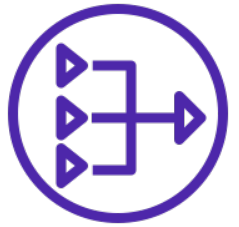
NAT Instance



NAT Instance – Comments

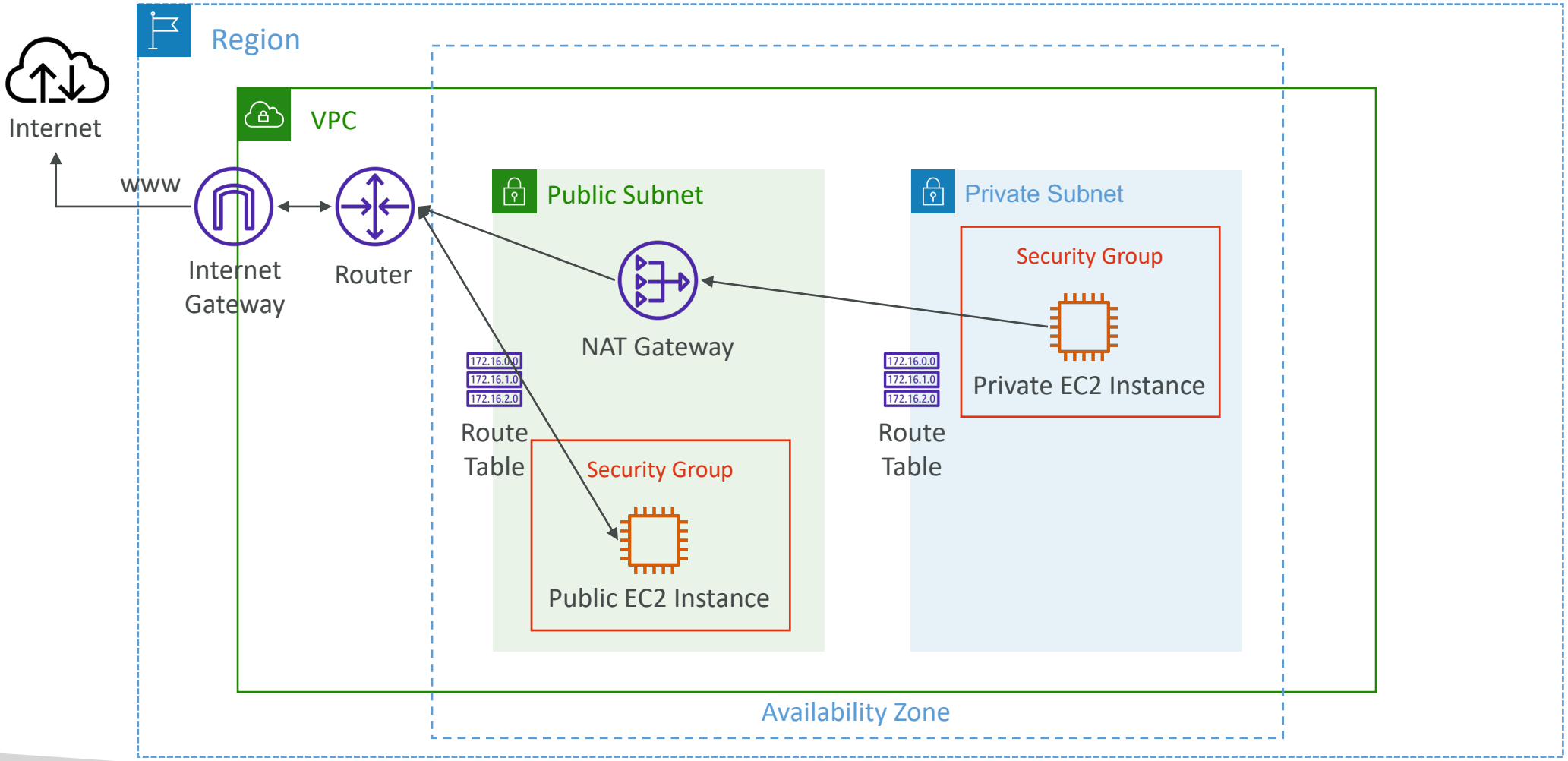
- Pre-configured Amazon Linux AMI is available
  - Reached the end of standard support on December 31, 2020
- Not highly available / resilient setup out of the box
  - You need to create an ASG in multi-AZ + resilient user-data script
- Internet traffic bandwidth depends on EC2 instance type
- You must manage Security Groups & rules:
  - Inbound:
    - Allow HTTP / HTTPS traffic coming from Private Subnets
    - Allow SSH from your home network (access is provided through Internet Gateway)
  - Outbound:
    - Allow HTTP / HTTPS traffic to the Internet

NAT Gateway



- AWS-managed NAT, higher bandwidth, high availability, no administration
- Pay per hour for usage and bandwidth
- NATGW is created in a specific Availability Zone, uses an Elastic IP
- Can't be used by EC2 instance in the same subnet (only from other subnets)
- Requires an IGW (Private Subnet => NATGW => IGW)
- 5 Gbps of bandwidth with automatic scaling up to 100 Gbps
- No Security Groups to manage / required

NAT Gateway

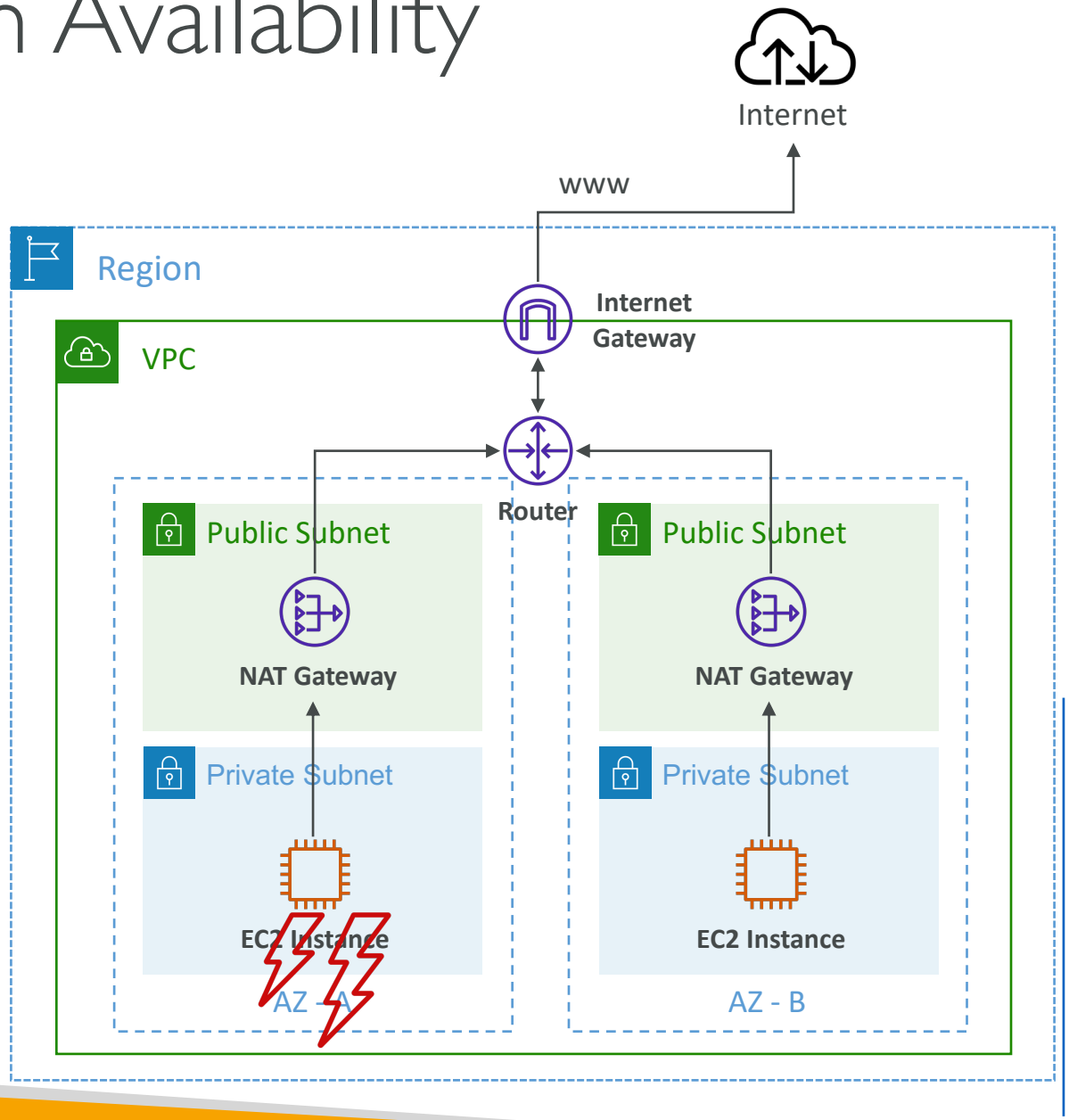

Region
Internet
VPC
www
Internet Gateway
Router
Public Subnet
NAT Gateway
172.16.0.0
172.16.1.0
172.16.2.0
Route Table
Security Group
Public EC2 Instance
Private Subnet
Security Group
Private EC2 Instance
172.16.0.0
172.16.1.0
172.16.2.0
Route Table
Availability Zone

NAT Gateway with High Availability

- **NAT Gateway is resilient within a single Availability Zone**

- Must create **multiple NAT Gateways** in **multiple AZs** for fault-tolerance

- There is no cross-AZ failover needed because if an AZ goes down it doesn't need NAT



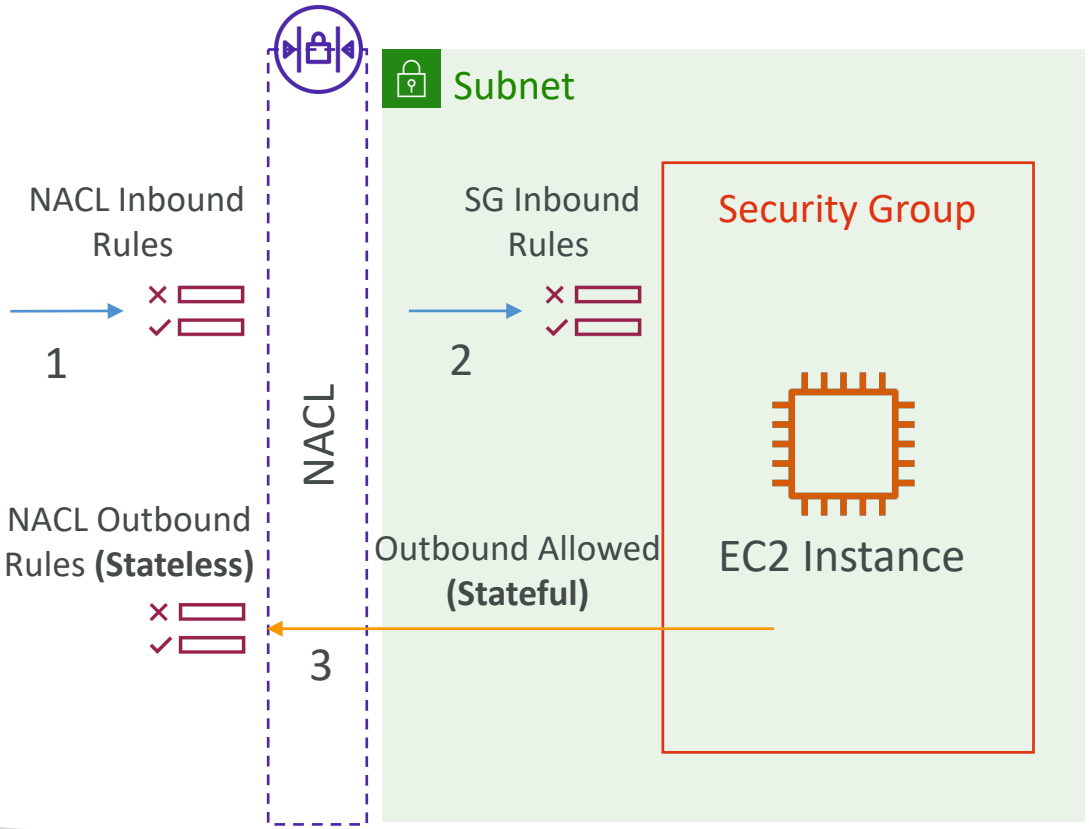
NAT Gateway vs. NAT Instance

	NAT Gateway	NAT Instance
Availability	Highly available within AZ (create in another AZ)	Use a script to manage failover between instances
Bandwidth	Up to 100 Gbps	Depends on EC2 instance type
Maintenance	Managed by AWS	Managed by you (e.g., software, OS patches, …)
Cost	Per hour & amount of data transferred	Per hour, EC2 instance type and size, + network $
Public IPv4	✓	✓
Private IPv4	✓	✓
Security Groups	✗	✓
Use as Bastion Host?	✗	✓

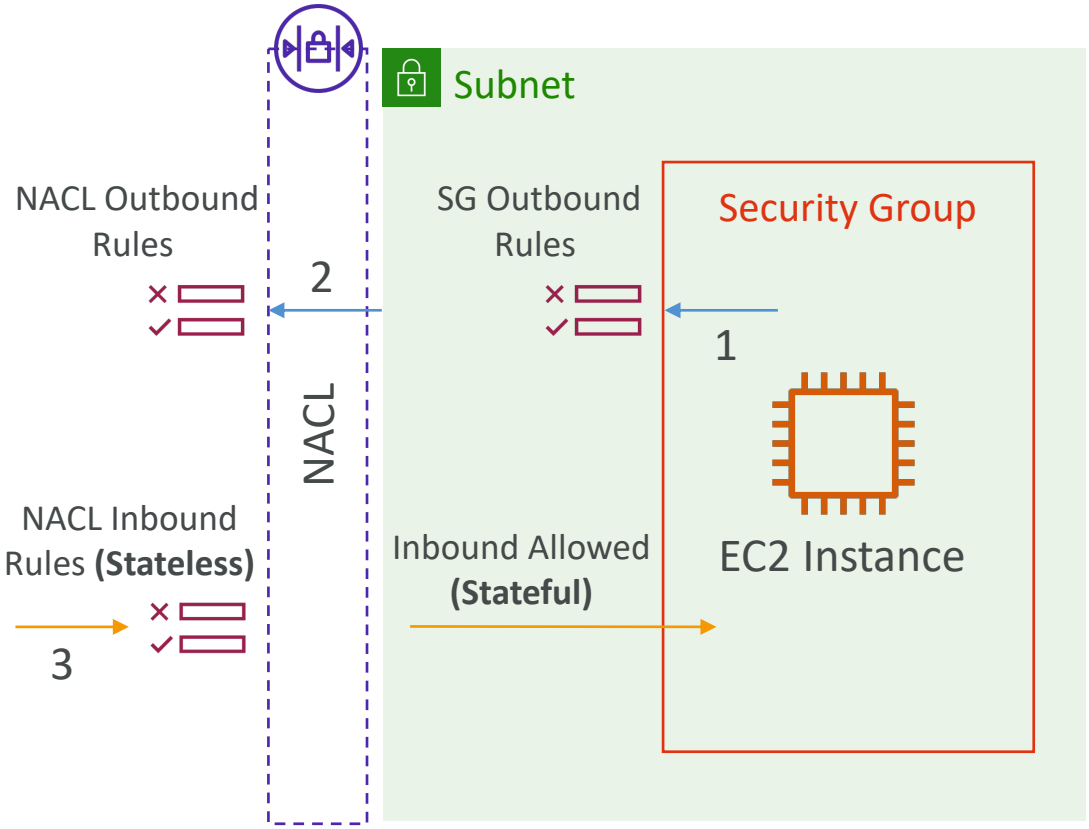
More at: https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-comparison.html

Security Groups & NACLs

Incoming Request



Outgoing Request

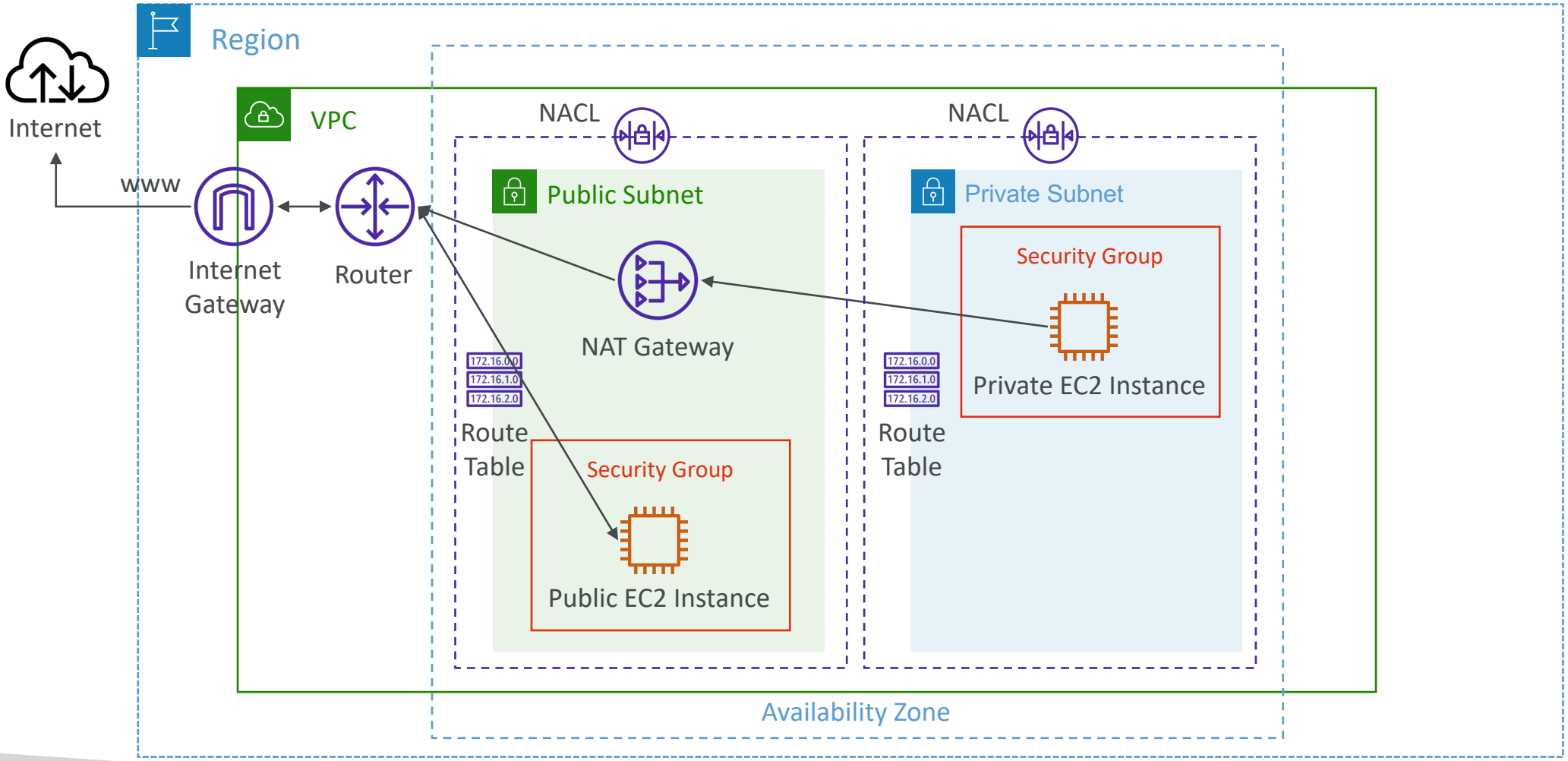


Network Access Control List (NACL)



- NACL are like a firewall which control traffic from and to **subnets**
- **One NACL per subnet,** new subnets are assigned the **Default NACL**
- You define **NACL Rules:**
  - Rules have a number (1-32766), higher precedence with a lower number
  - First rule match will drive the decision
  - Example: if you define #100 ALLOW 10.0.0.10/32 and #200 DENY 10.0.0.10/32, the IP address will be allowed because 100 has a higher precedence over 200
  - The last rule is an asterisk (*) and denies a request in case of no rule match
  - AWS recommends adding rules by increment of 100
- Newly created NACLs will deny everything
- NACL are a great way of blocking a specific IP address at the subnet level

NACLs


Region
Internet
VPC
NACL
NACL
www
Public Subnet
Private Subnet
Security Group
Internet Gateway
Router
NAT Gateway
172.16.0.0
172.16.1.0
172.16.2.0
Private EC2 Instance
Route Table
Route Table
Security Group
Public EC2 Instance
Availability Zone

Default NACL

- Accepts everything inbound/outbound with the subnets it's associated with
- Do **NOT** modify the Default NACL, instead create custom NACLs



Default NACL for a VPC that supports IPv4

Inbound Rules

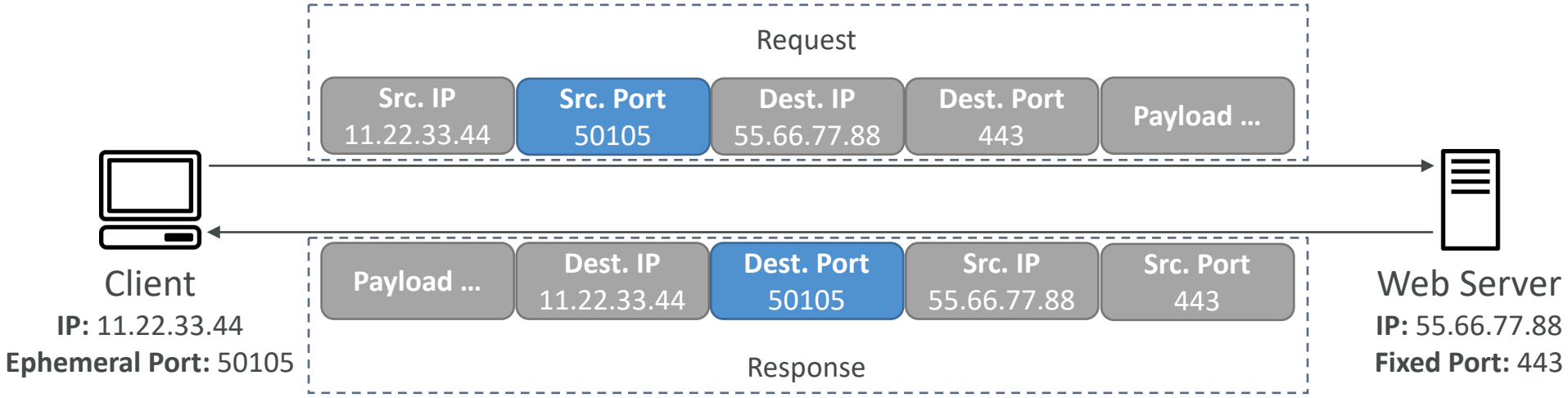
Rule #	Type	Protocol	Port Range	Source	Allow/Deny
100	All IPv4 Traffic	All	All	0.0.0.0/0	ALLOW
*	All IPv4 Traffic	All	All	0.0.0.0/0	DENY

Outbound Rules

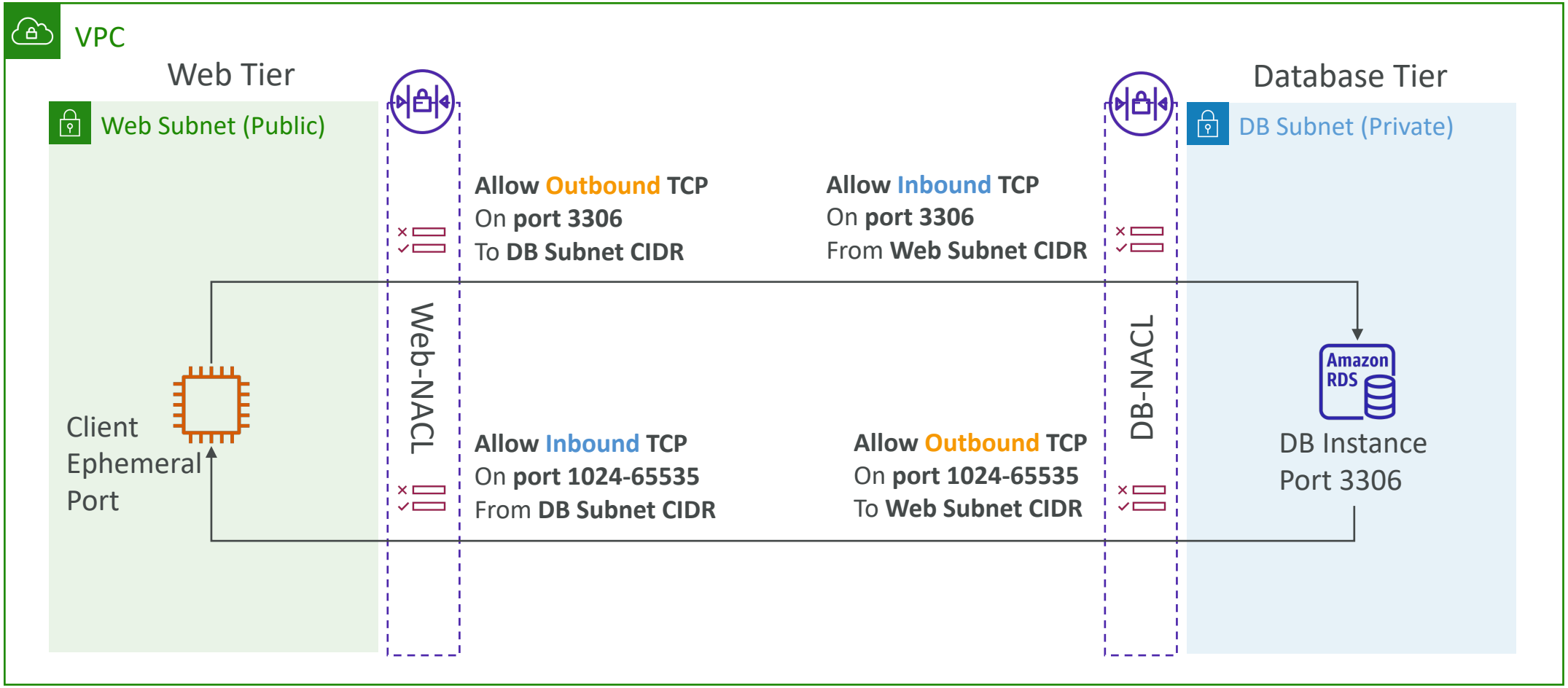
Rule #	Type	Protocol	Port Range	Destination	Allow/Deny
100	All IPv4 Traffic	All	All	0.0.0.0/0	ALLOW
*	All IPv4 Traffic	All	All	0.0.0.0/0	DENY

Ephemeral Ports

- For any two endpoints to establish a connection, they must use ports
- Clients connect to a **defined port**, and expect a response on an **ephemeral port**
- Different Operating Systems use different port ranges, examples:
  - IANA & MS Windows 10 ➔ 49152 – 65535
  - Many Linux Kernels ➔ 32768 – 60999

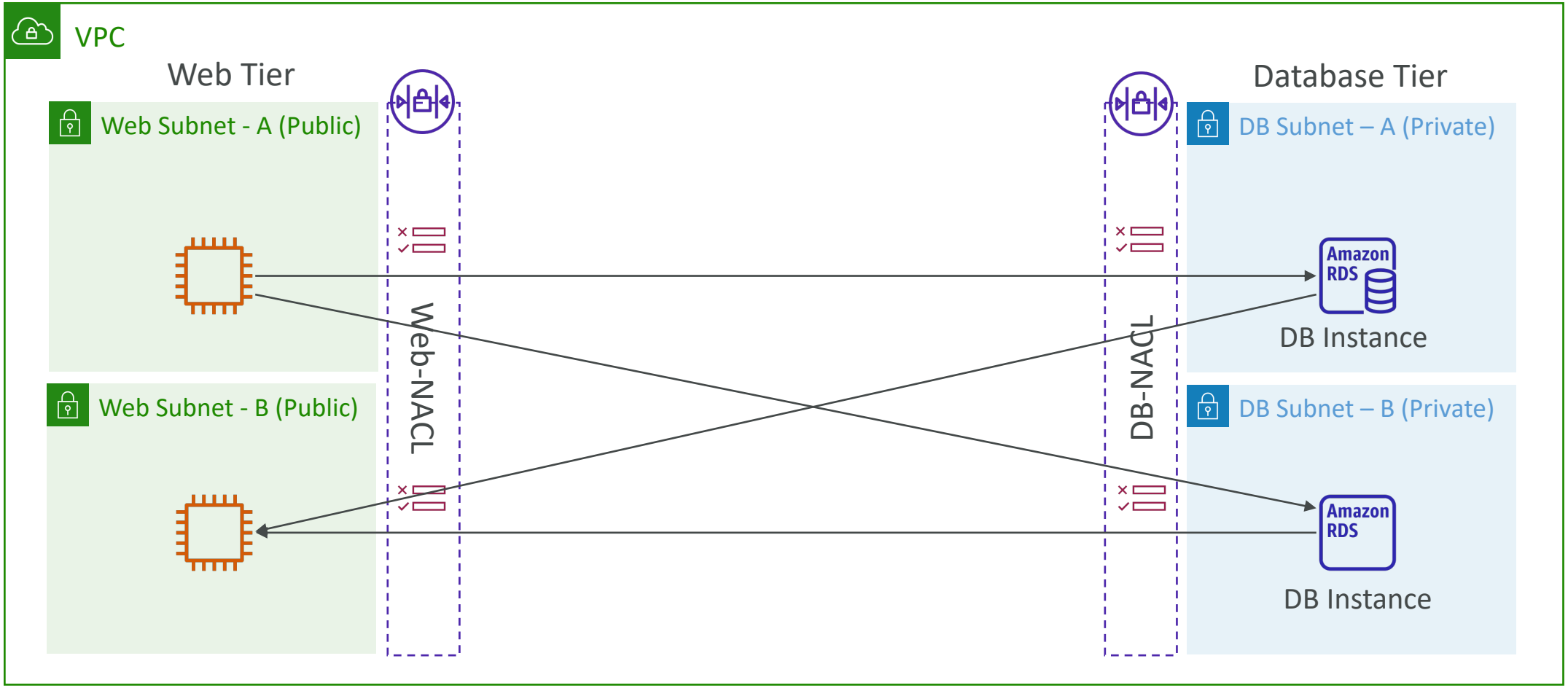


NACL with Ephemeral Ports



https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html#nacl-ephemeral-ports

Create NACL rules for each target subnets CIDR



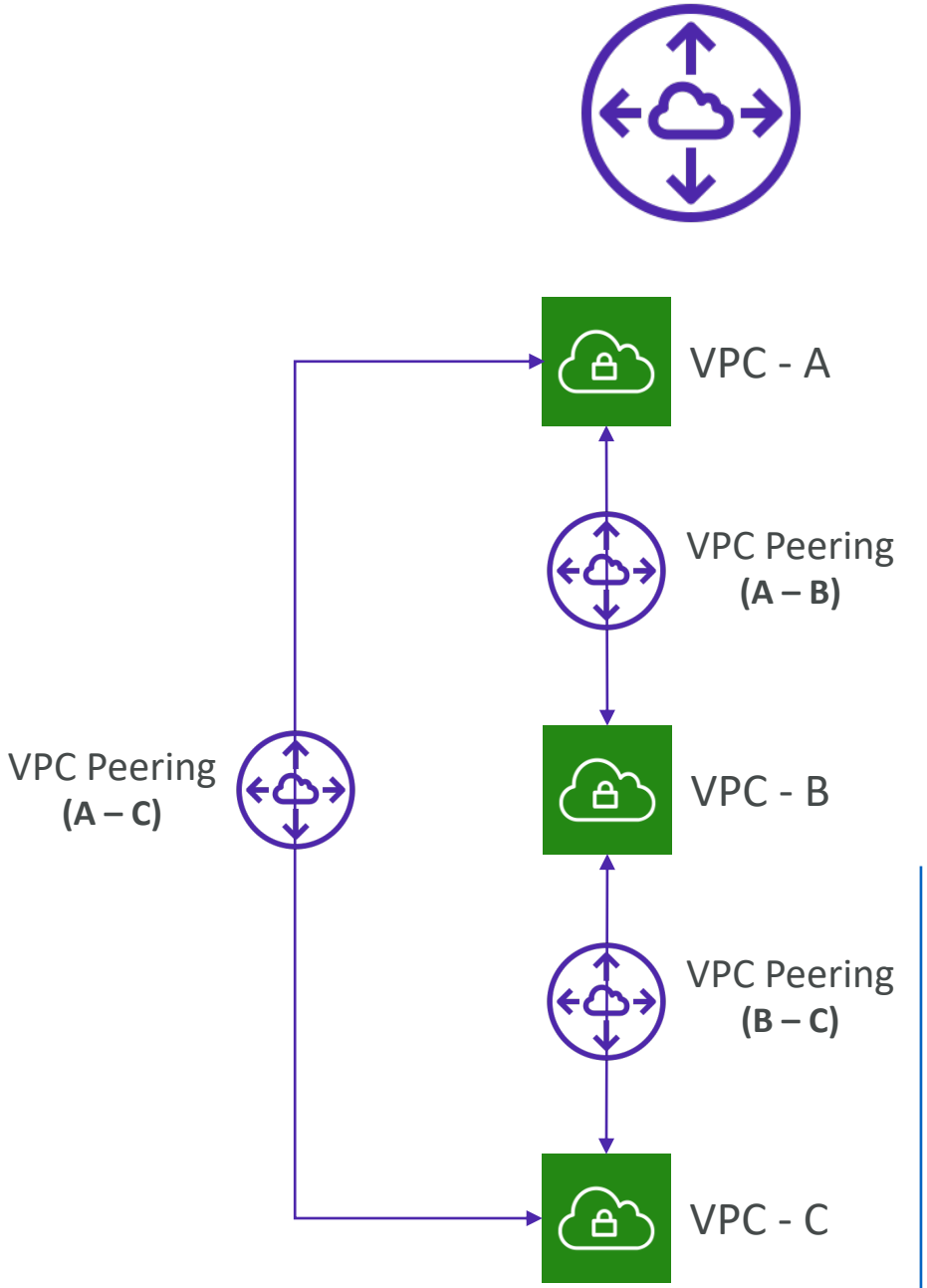
Security Group vs. NACLs

Security Group	NACL
Operates at the instance level	Operates at the subnet level
Supports allow rules only	Supports allow rules and deny rules
Stateful: return traffic is automatically allowed, regardless of any rules	**Stateless:** return traffic must be explicitly allowed by rules (think of ephemeral ports)
All rules are evaluated before deciding whether to allow traffic	Rules are evaluated in order (lowest to highest) when deciding whether to allow traffic, first match wins
Applies to an EC2 instance when specified by someone	Automatically applies to all EC2 instances in the subnet that it's associated with

NACL Examples: https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html

VPC Peering

- Privately connect two VPCs using AWS' network
- Make them behave as if they were in the same network
- Must not have overlapping CIDRs
- VPC Peering connection is **NOT transitive** (must be established for each VPC that need to communicate with one another)
- **You must update route tables in each VPC's subnets to ensure EC2 instances can communicate with each other**



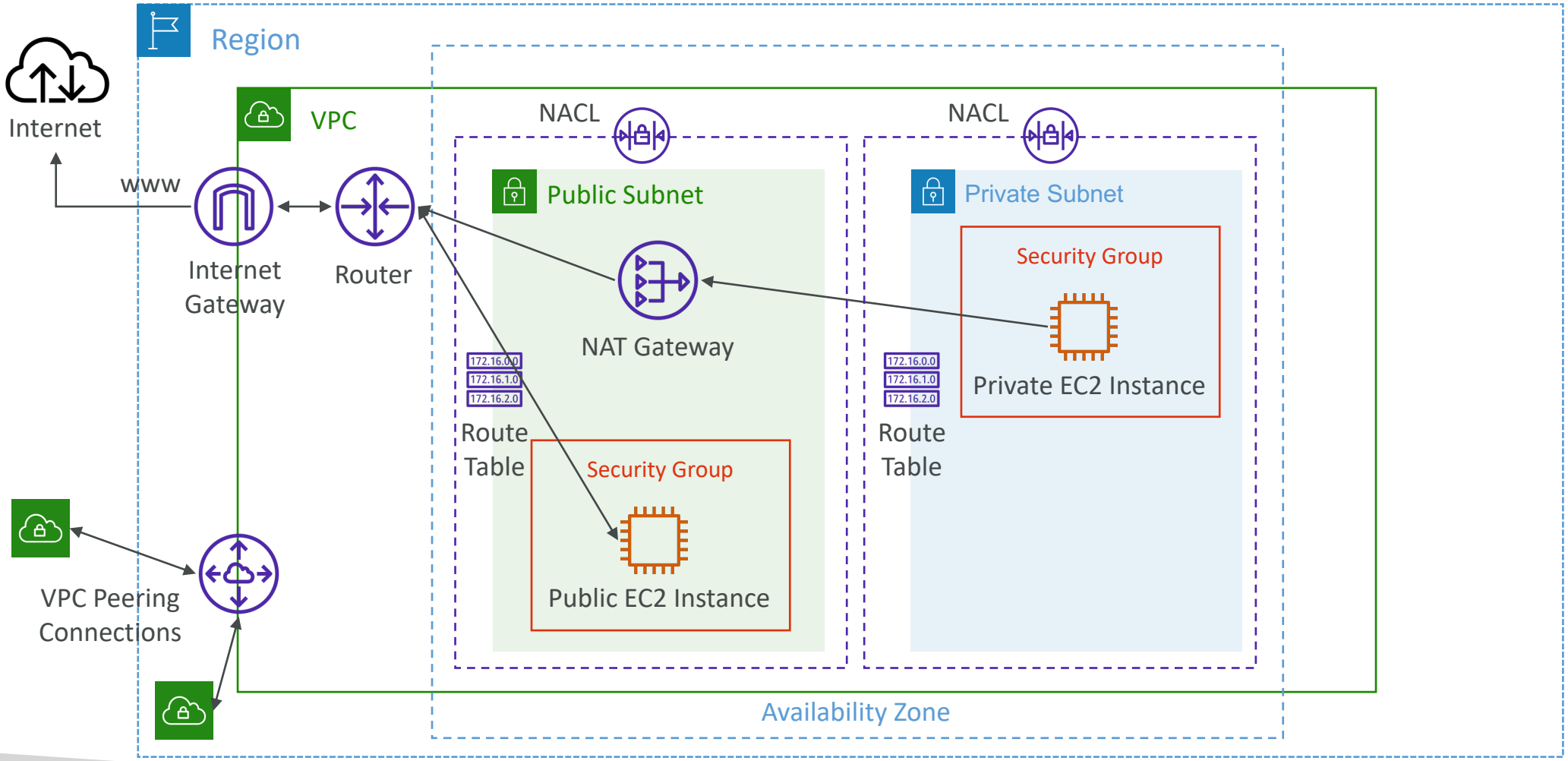
VPC Peering – Good to know

- You can create VPC Peering connection between VPCs in **different AWS accounts/regions**
- You can reference a security group in a peered VPC (works cross accounts – same region)

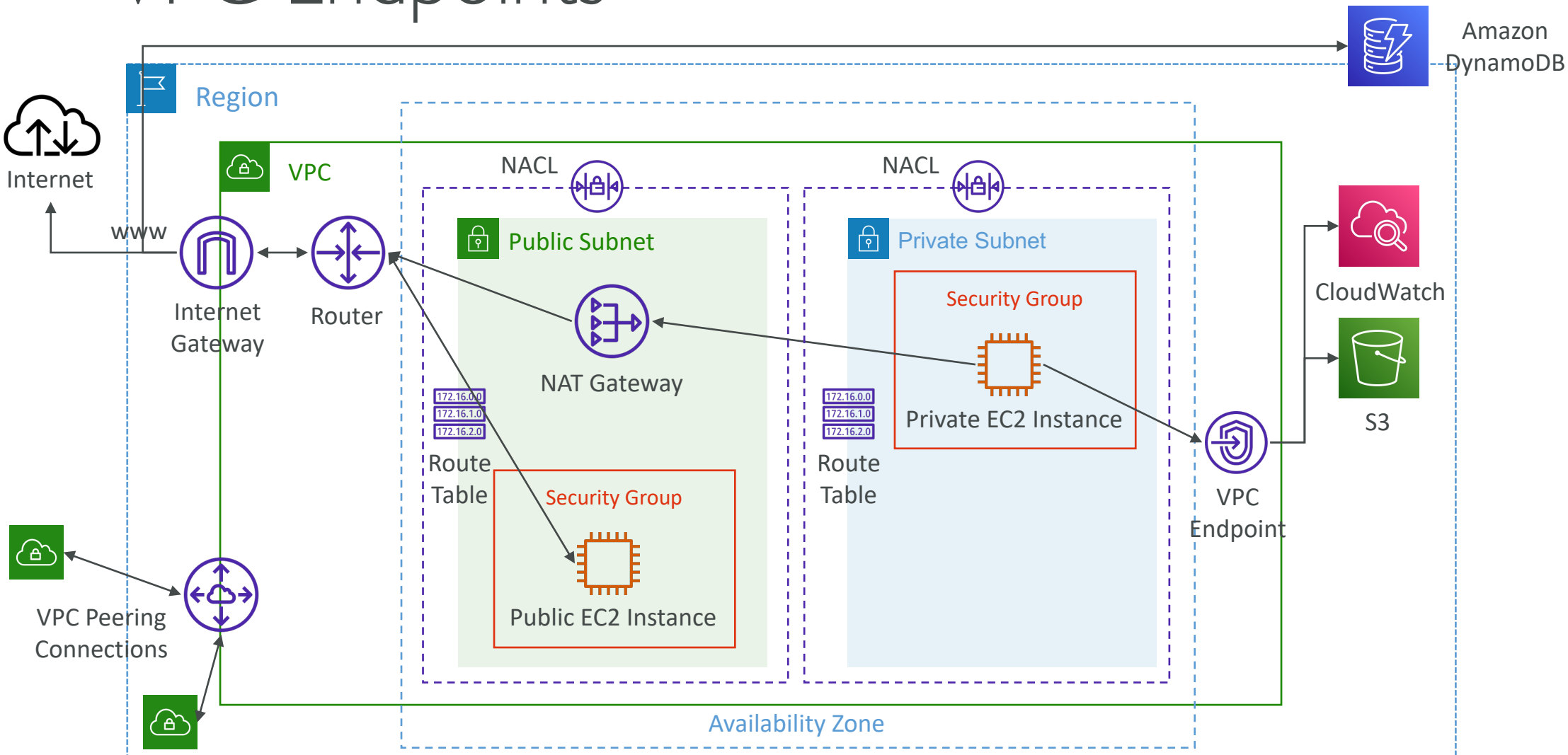
Type	Protocol	Port range	Source
HTTP	TCP	80	sg-04991f9af3473b939 / default
HTTP	TCP	80	[redacted] / sg-027ad1f7865d4be76

Account ID

VPC Peering


Region
Internet
VPC
NACL
NACL
www
Public Subnet
Private Subnet
Internet Gateway
Router
Security Group
NAT Gateway
172.16.0.0
172.16.1.0
172.16.2.0
Private EC2 Instance
Route Table
Route Table
Security Group
Public EC2 Instance
VPC Peering Connections
Availability Zone

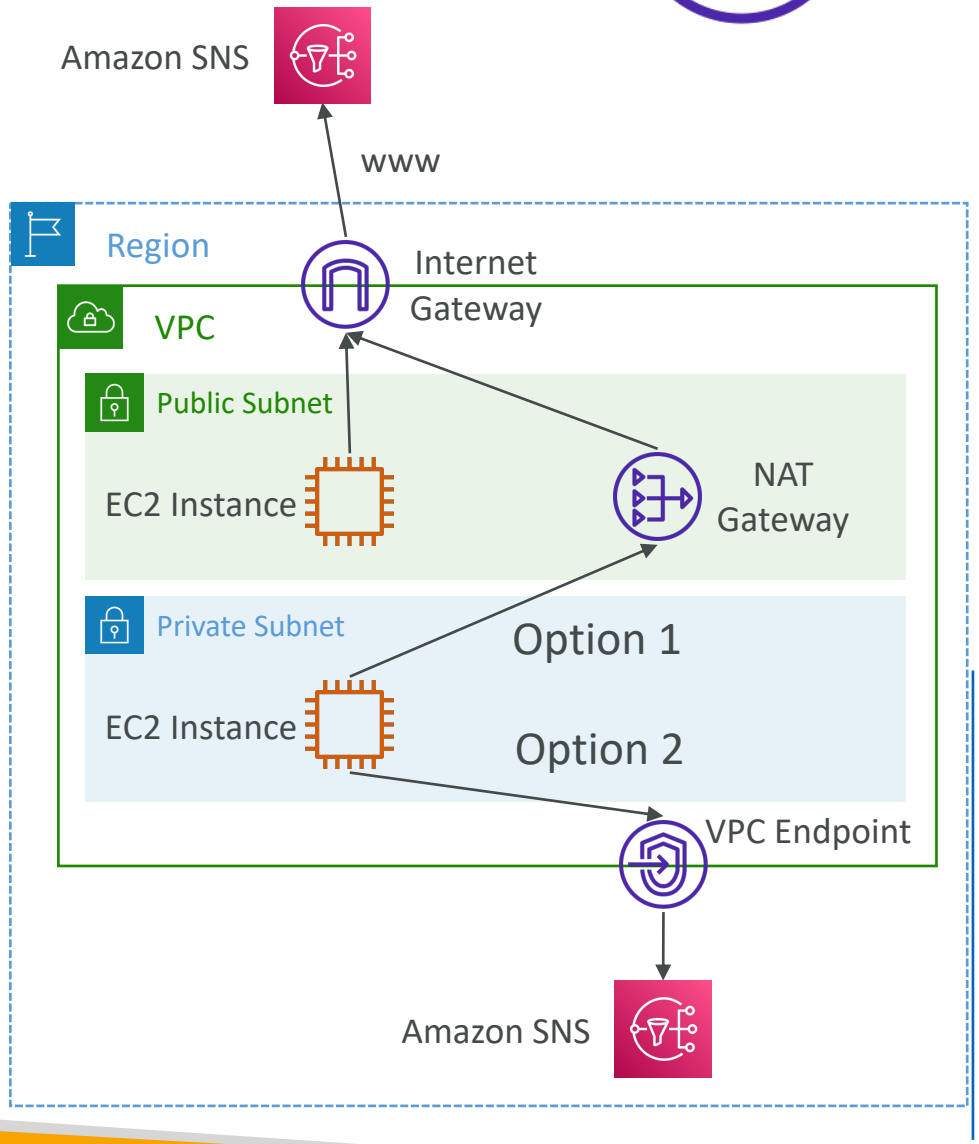
VPC Endpoints



VPC Endpoints (AWS PrivateLink)

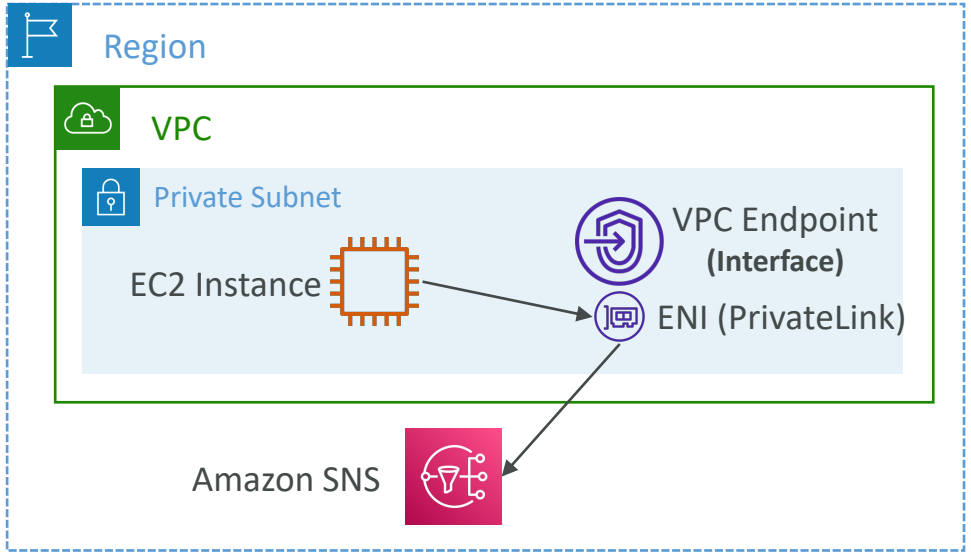


- Every AWS service is publicly exposed (public URL)
- VPC Endpoints (powered by AWS PrivateLink) allows you to connect to AWS services using a **private network** instead of using the public Internet
- They're redundant and scale horizontally
- They remove the need of IGW, NATGW, … to access AWS Services
- In case of issues:
  - Check DNS Setting Resolution in your VPC
  - Check Route Tables

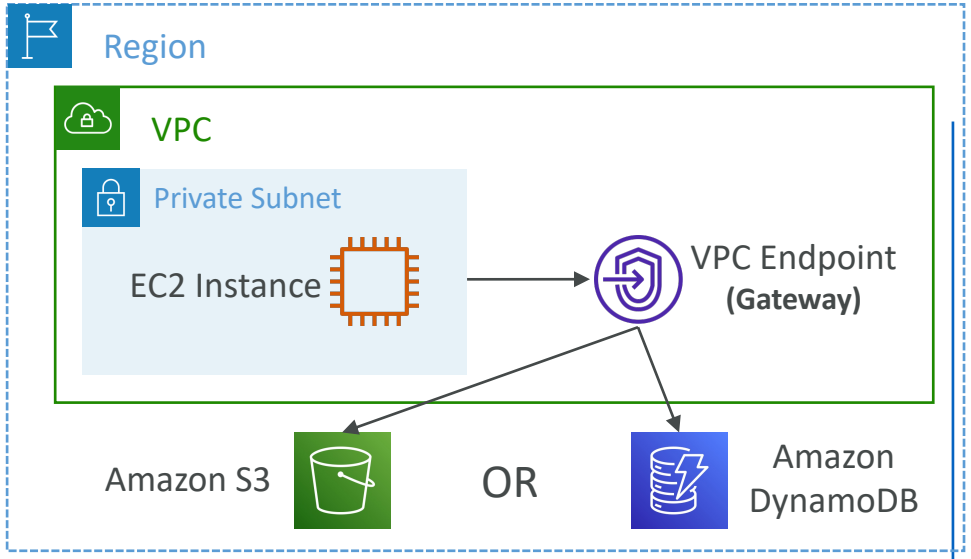


Types of Endpoints

- **Interface Endpoints (powered by PrivateLink)**
  - Provisions an ENI (private IP address) as an entry point (must attach a Security Group)
  - Supports most AWS services
  - $ per hour + $ per GB of data processed

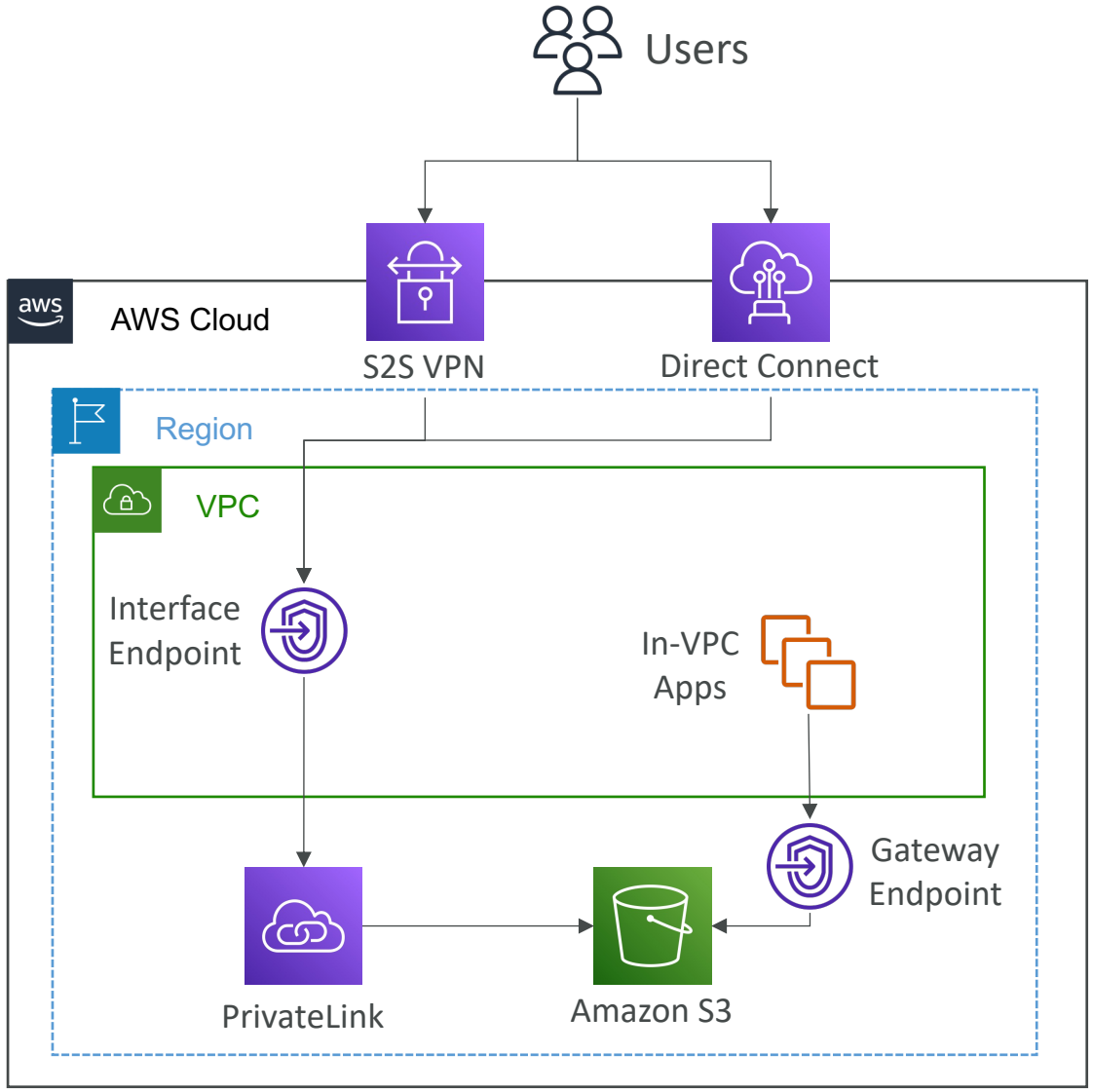


- **Gateway Endpoints**
  - Provisions a gateway and must be used as a target in a route table (does not use security groups)
  - Supports both S3 and DynamoDB
  - Free



Gateway or Interface Endpoint for S3?

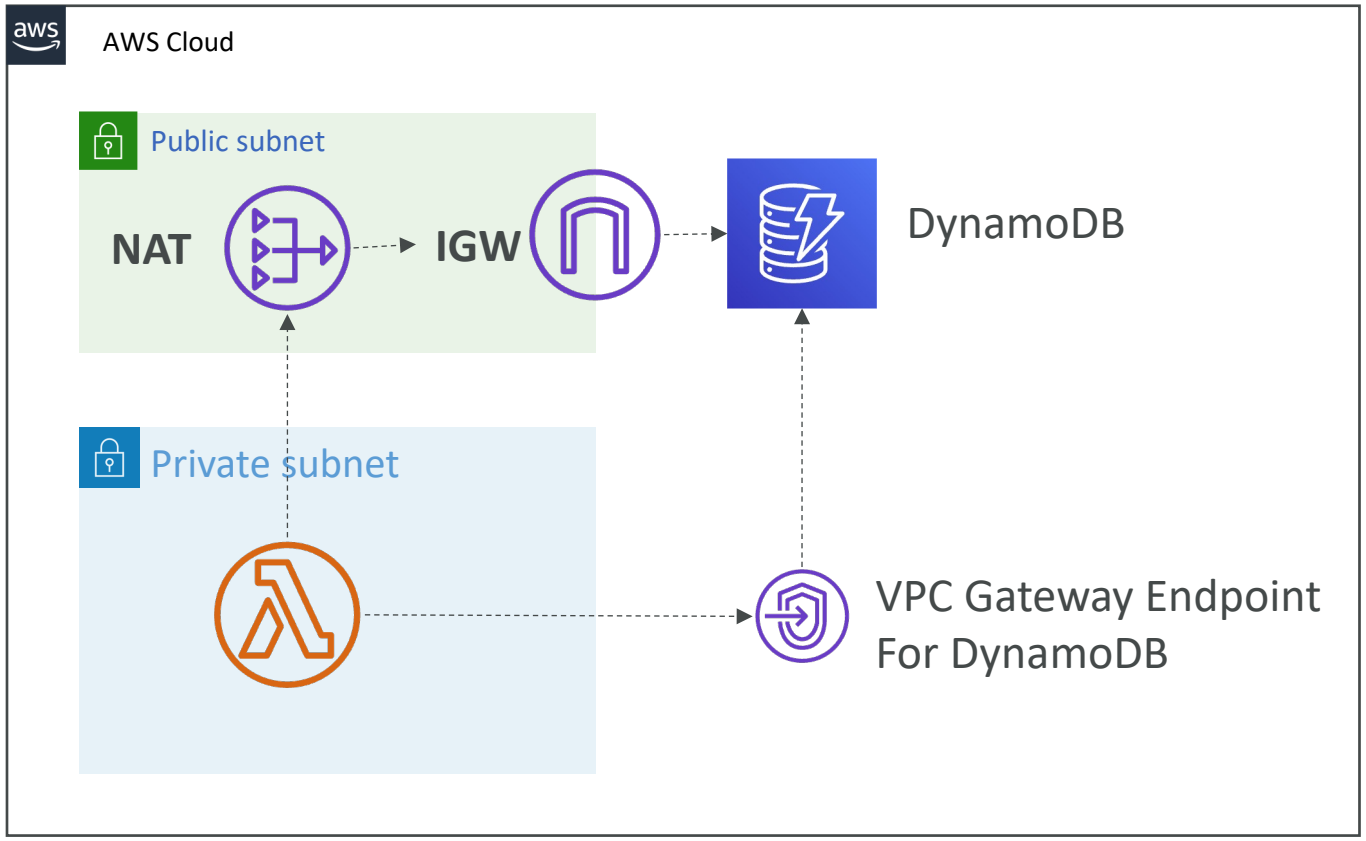
- **Gateway is most likely going to be preferred all the time at the exam**
- Cost: free for Gateway, $ for interface endpoint
- Interface Endpoint is preferred access is required from on-premises (Site to Site VPN or Direct Connect), a different VPC or a different region



Lambda in VPC accessing DynamoDB

- DynamoDB is a public service from AWS
- Option 1: Access from the public internet
  - Because Lambda is in a VPC, it needs a NAT Gateway in a public subnet and an internet gateway

- **Option 2 (better & free): Access from the private VPC network**
  - Deploy a VPC Gateway endpoint for DynamoDB
  - Change the Route Tables

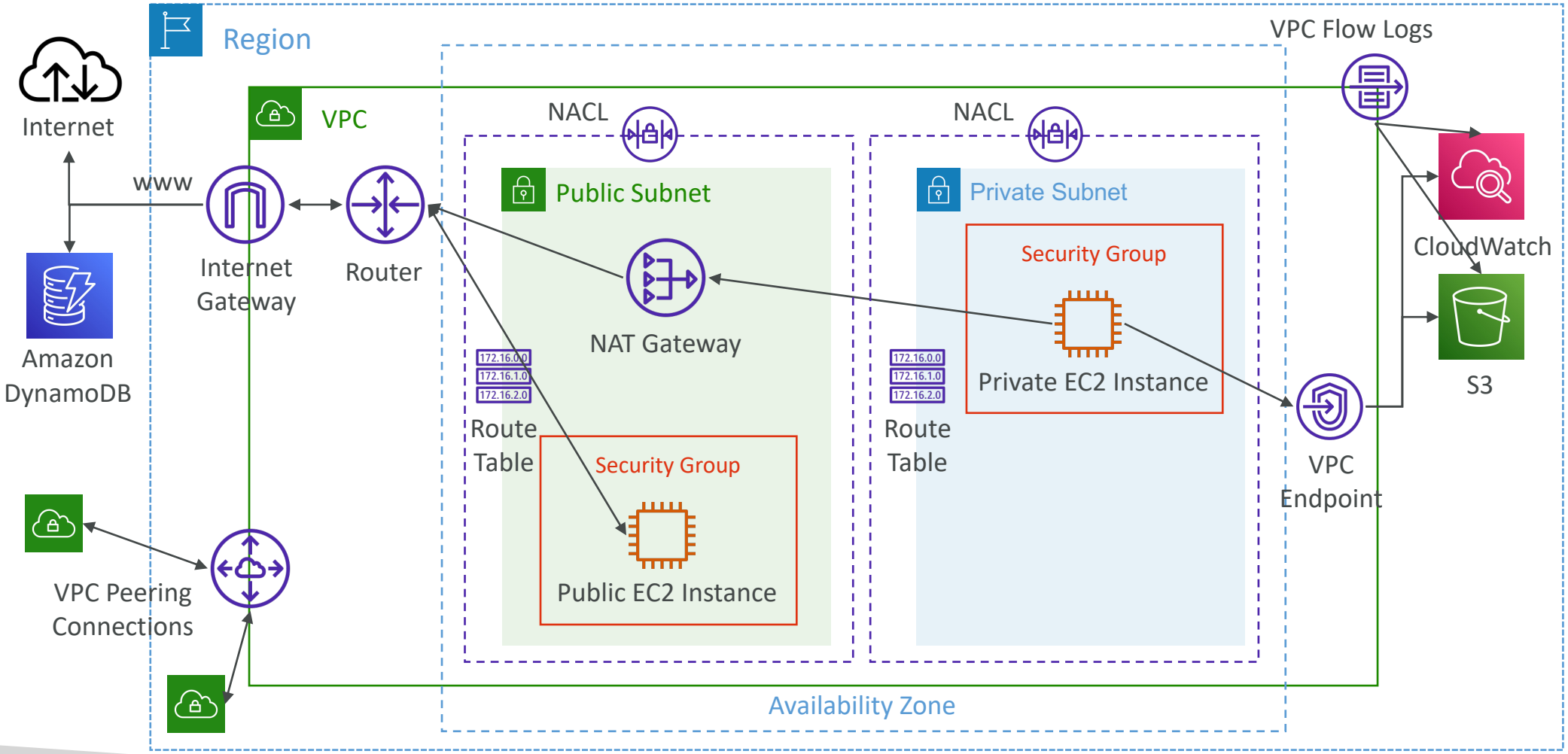


VPC Flow Logs

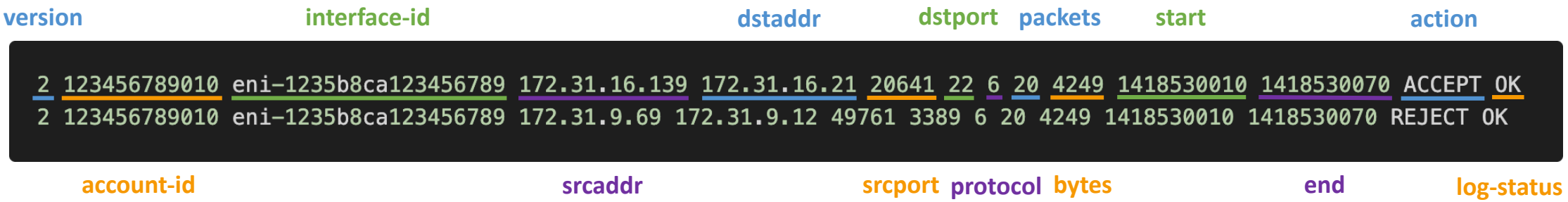


- Capture information about IP traffic going into your interfaces:
  - VPC Flow Logs
  - Subnet Flow Logs
  - Elastic Network Interface (ENI) Flow Logs
- Helps to monitor & troubleshoot connectivity issues
- Flow logs data can go to S3, CloudWatch Logs, and Kinesis Data Firehose
- Captures network information from AWS managed interfaces too: ELB, RDS, ElastiCache, Redshift, WorkSpaces, NATGW, Transit Gateway…

VPC Flow Logs



VPC Flow Logs Syntax



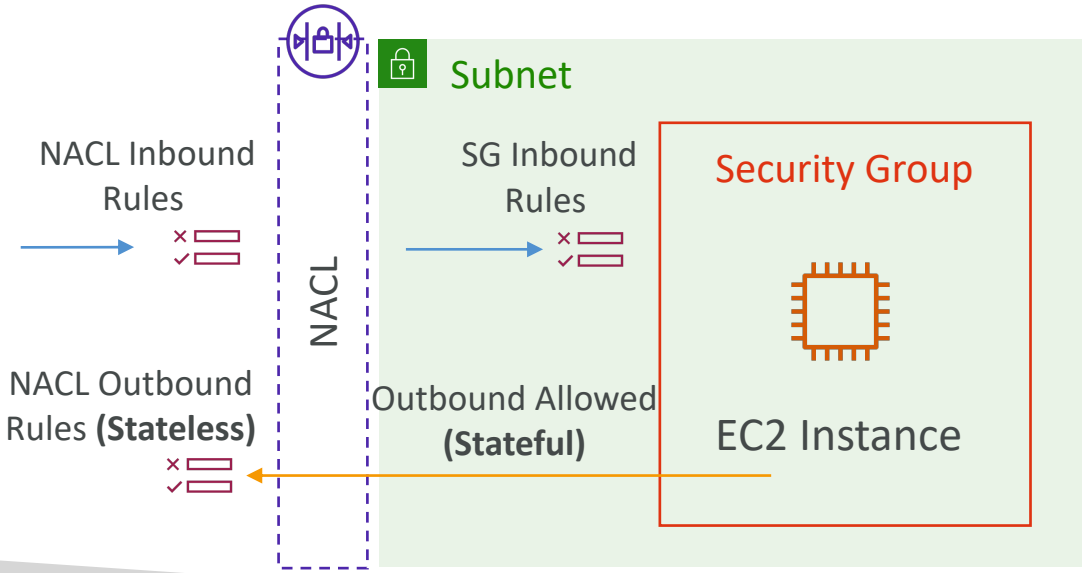
- **srcaddr & dstaddr** – help identify problematic IP
- **srcport & dstport** – help identity problematic ports
- **Action** – success or failure of the request due to Security Group / NACL
- Can be used for analytics on usage patterns, or malicious behavior
- **Query VPC flow logs using Athena on S3 or CloudWatch Logs Insights**
- Flow Logs examples: https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs-records-examples.html

VPC Flow Logs – Troubleshoot SG & NACL issues

Look at the **"ACTION"** field

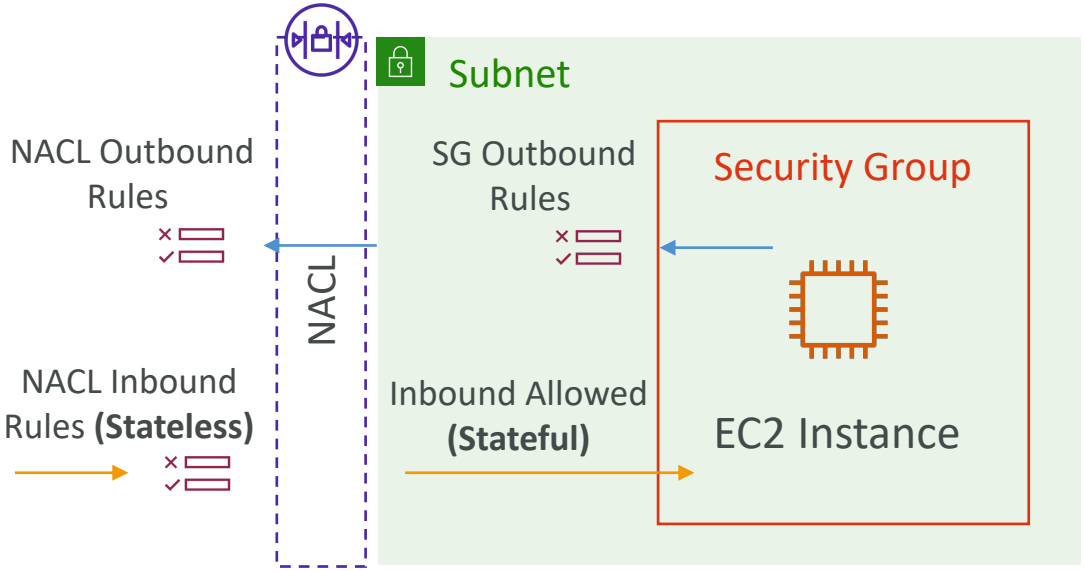
Incoming Requests

- Inbound REJECT => NACL or SG
- Inbound ACCEPT, Outbound REJECT => NACL

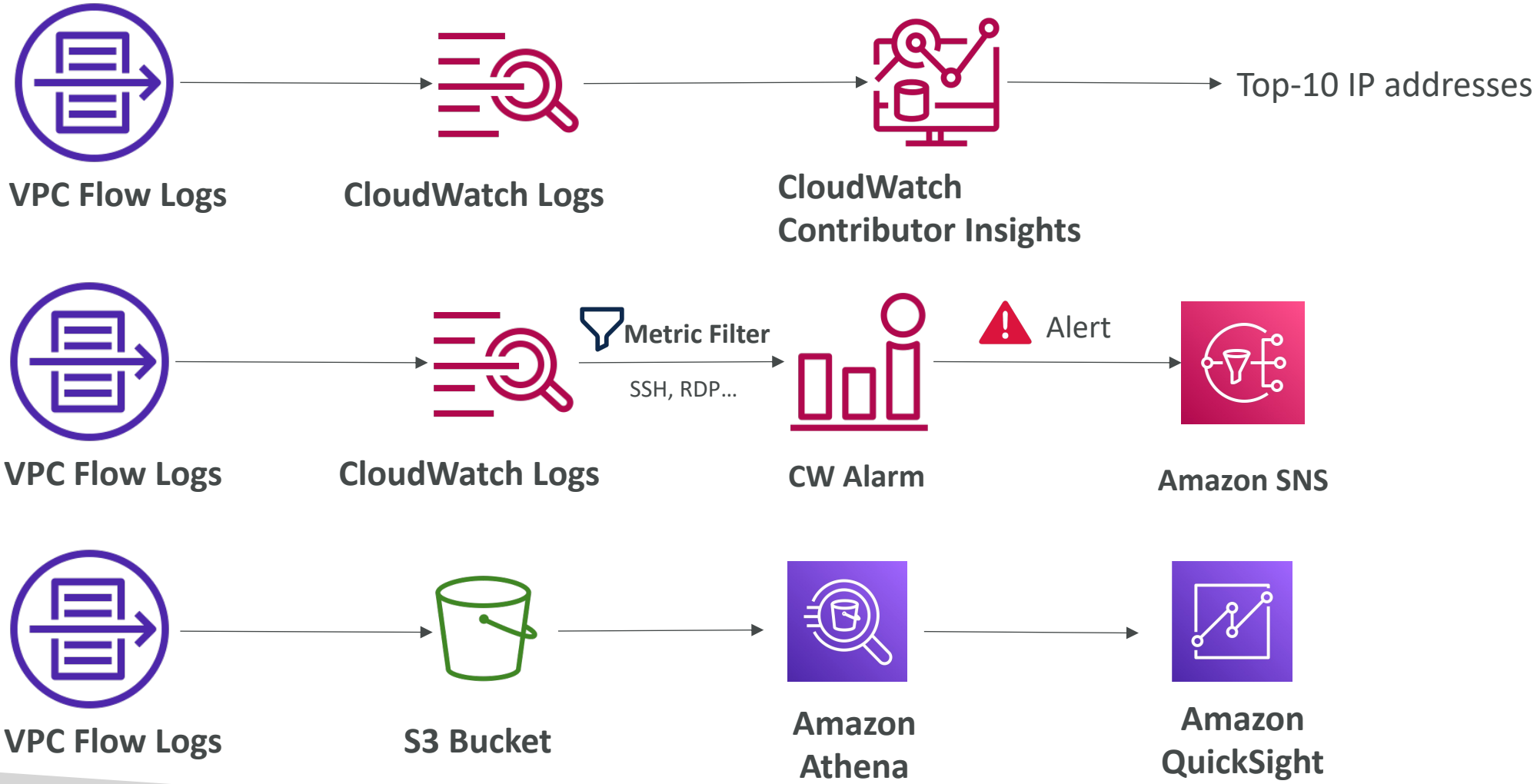


Outgoing Requests

- Outbound REJECT => NACL or SG
- Outbound ACCEPT, Inbound REJECT => NACL

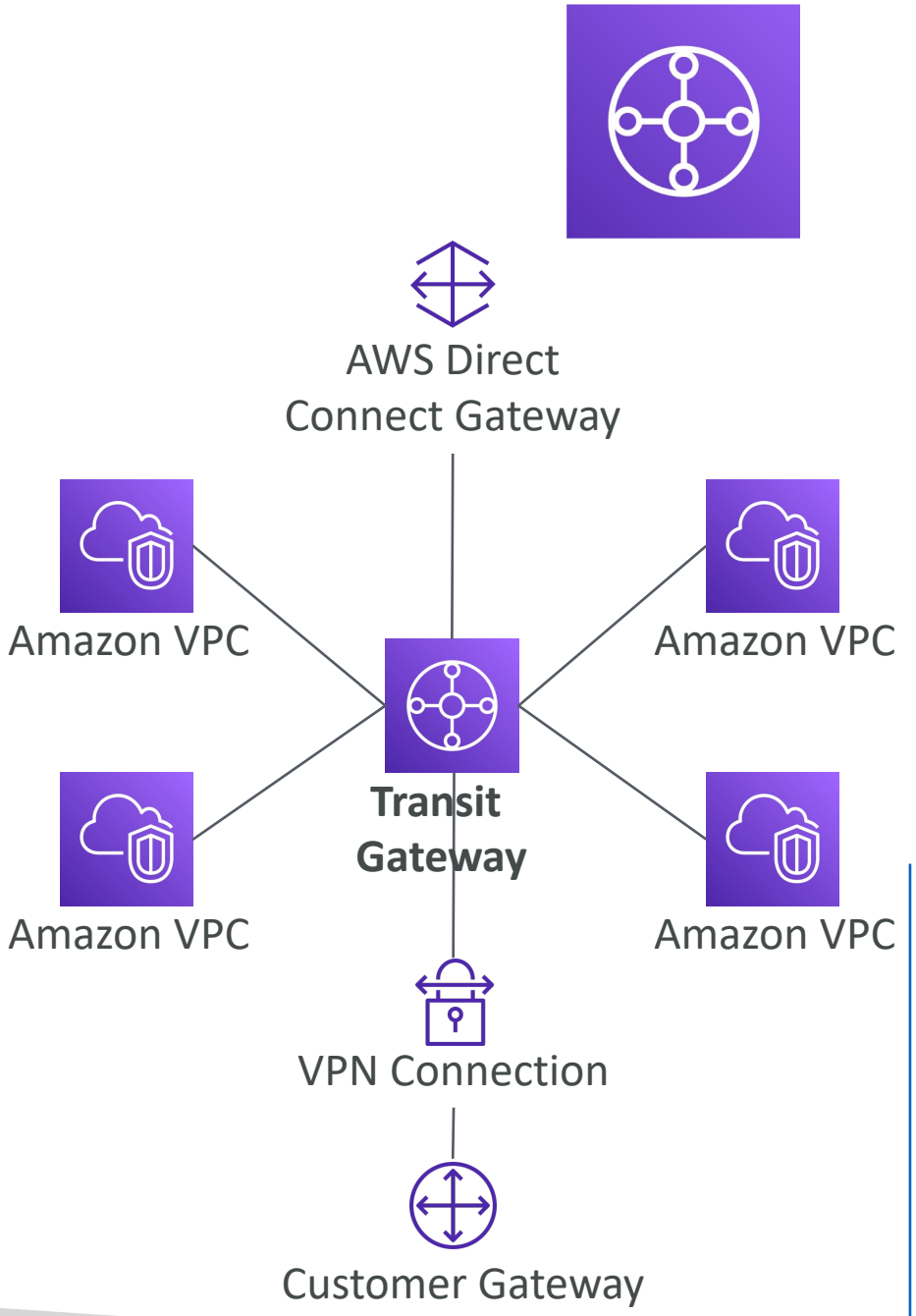


VPC Flow Logs – Architectures

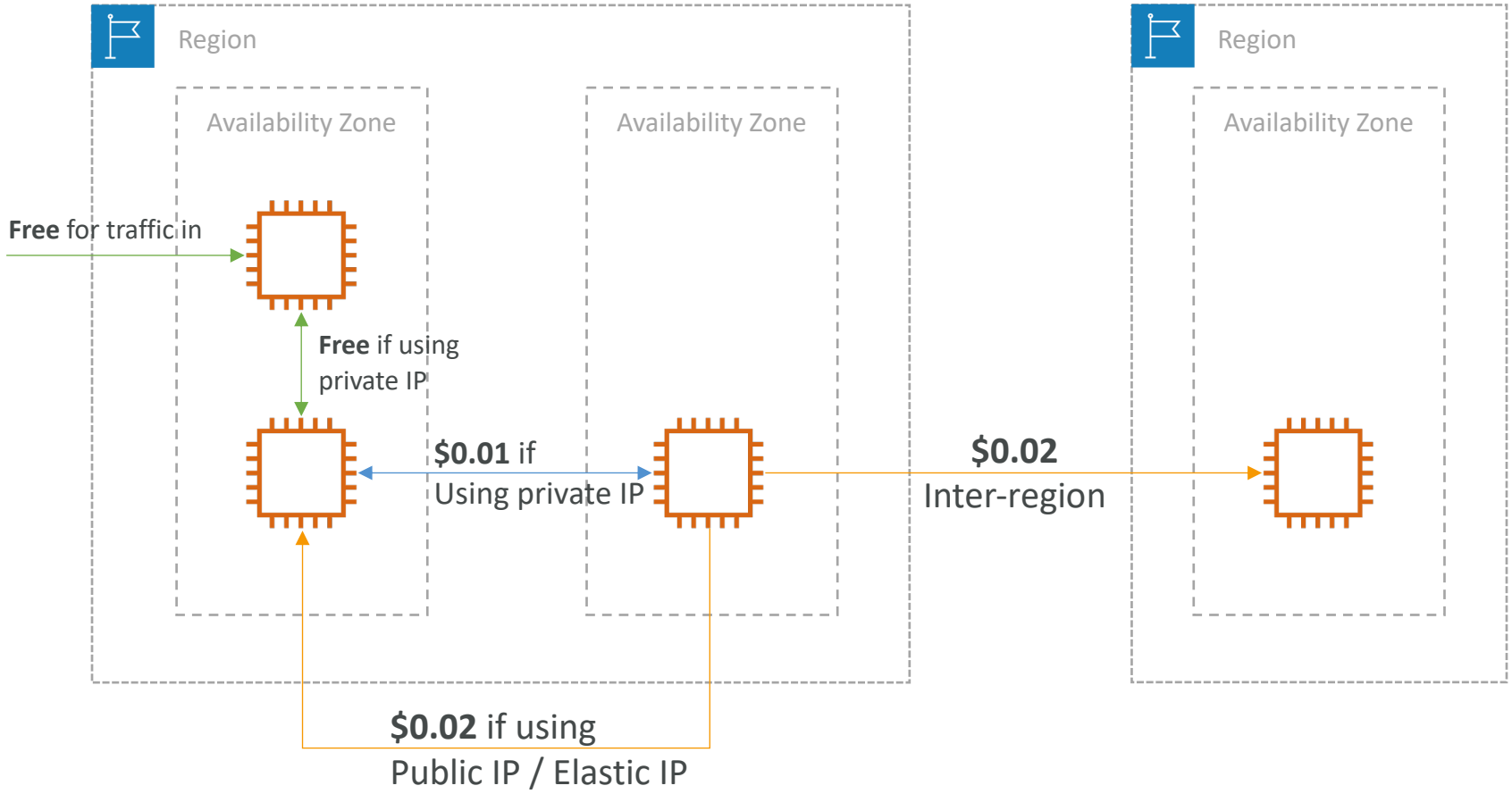


Transit Gateway

- **For having transitive peering between thousands of VPC and on-premises, hub-and-spoke (star) connection**
- Regional resource, can work cross-region
- Share cross-account using Resource Access Manager (RAM)
- You can peer Transit Gateways across regions
- Route Tables: limit which VPC can talk with other VPC
- Works with Direct Connect Gateway, VPN connections
- Supports **IP Multicast** (not supported by any other AWS service)



Networking Costs in AWS per GB - Simplified

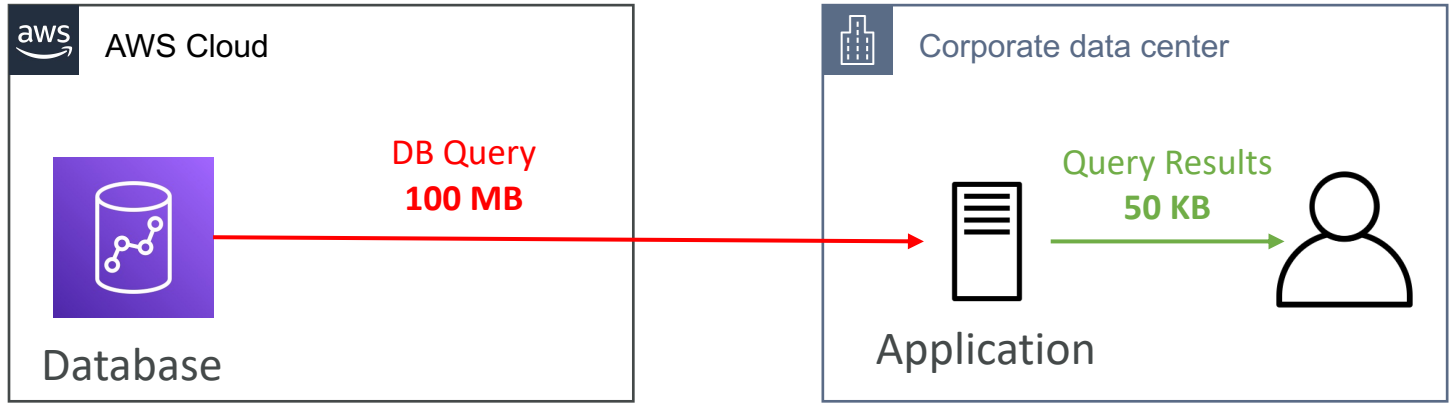


- Use Private IP instead of Public IP for good savings and better network performance
- Use same AZ for maximum savings (at the cost of high availability)

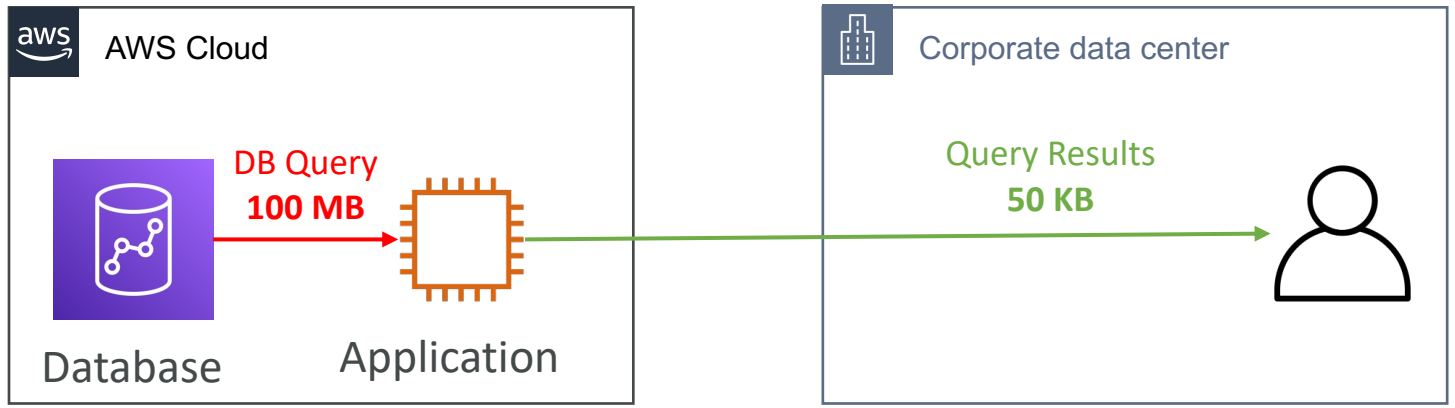
Minimizing egress traffic network cost

- Egress traffic: outbound traffic (from AWS to outside)
- Ingress traffic: inbound traffic - from outside to AWS (typically free)
- Try to keep as much internet traffic within AWS to minimize costs
- **Direct Connect location that are co-located in the same AWS Region result in lower cost for egress network**

Egress cost is high

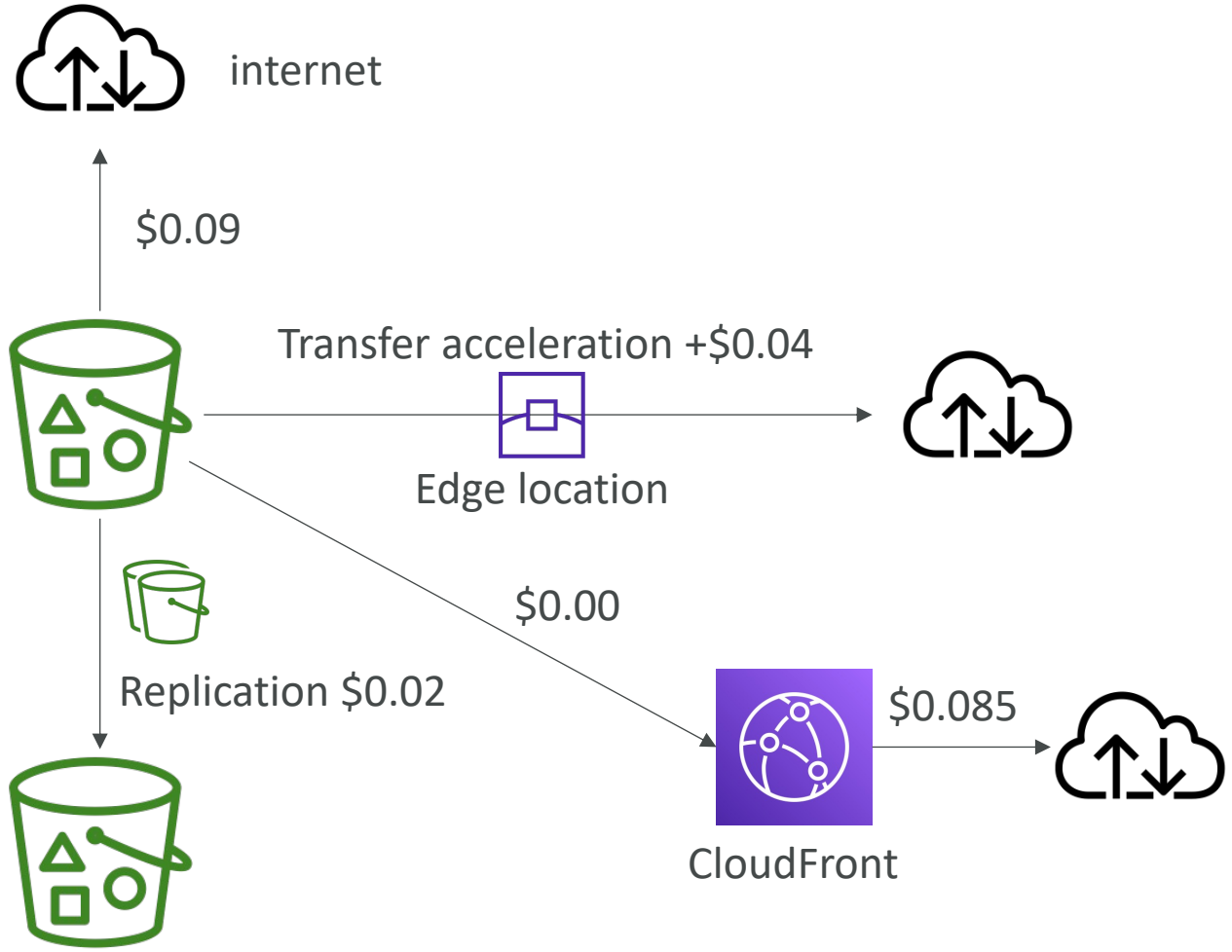


Egress cost is minimized



S3 Data Transfer Pricing – Analysis for USA

- **S3 ingress:** free
- **S3 to Internet:** $0.09 per GB
- **S3 Transfer Acceleration:**
  - Faster transfer times (50 to 500% better)
  - Additional cost on top of Data Transfer Pricing: +$0.04 to $0.08 per GB
- **S3 to CloudFront:** $0.00 per GB
- **CloudFront to Internet:** $0.085 per GB (slightly cheaper than S3)
  - Caching capability (lower latency)
  - Reduce costs associated with S3 Requests Pricing (7x cheaper with CloudFront)
- **S3 Cross Region Replication:** $0.02 per GB



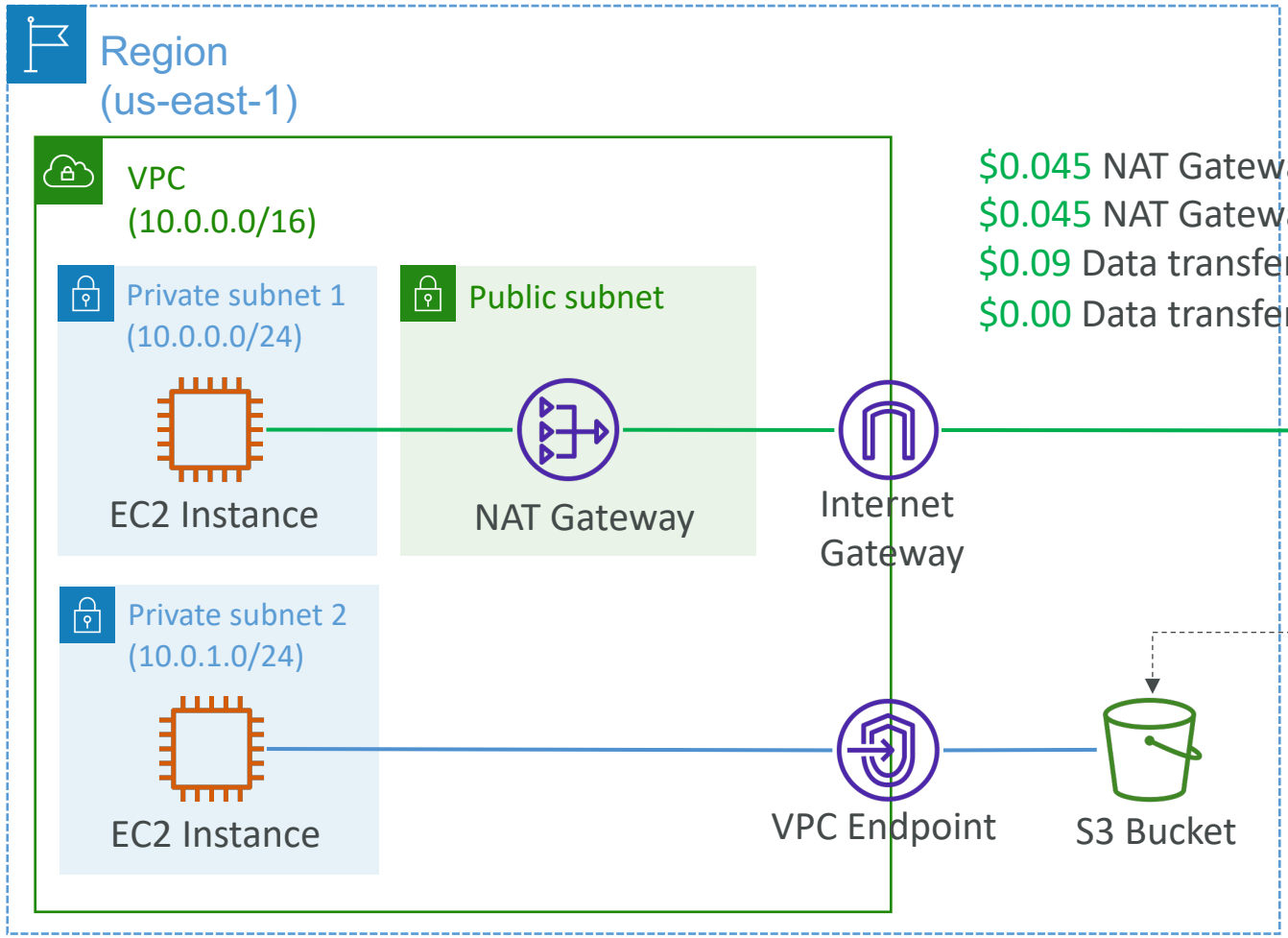
Pricing:
NAT Gateway vs Gateway VPC Endpoint

Subnet 1 route table

Destination	Target
10.0.0.0/16	Local
0.0.0.0/0	igw-id

Subnet 2 route table

Destination	Target
10.0.0.0/16	Local
pl-id for Amazon S3	vpce-id



$0.045 NAT Gateway / hour
$0.045 NAT Gateway data processed / GB
$0.09 Data transfer out to S3 (cross-region)
$0.00 Data transfer out to S3 (same-region)

No cost for using Gateway Endpoint.
$0.01 Data transfer in/out (same-region)

Network Protection on AWS

- To protect network on AWS, we've seen
  - Network Access Control Lists (NACLs)
  - Amazon VPC security groups
  - AWS WAF (protect against malicious requests)
  - AWS Shield & AWS Shield Advanced
  - AWS Firewall Manager (to manage them across accounts)

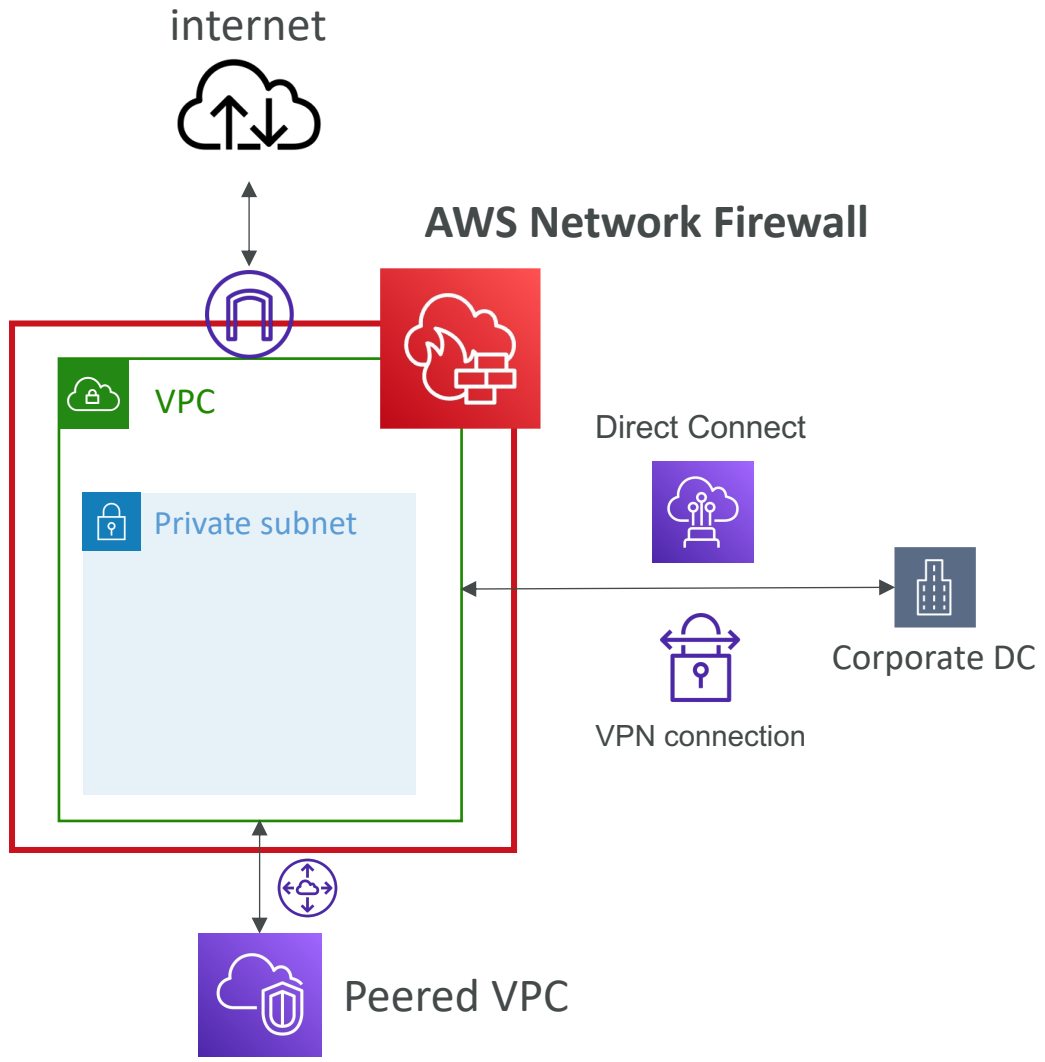
- But what if we want to protect in a sophisticated way our entire VPC?

AWS Network Firewall

- Protect your entire Amazon VPC

- From Layer 3 to Layer 7 protection
- Any direction, you can inspect
  - VPC to VPC traffic
  - Outbound to internet
  - Inbound from internet
  - To / from Direct Connect & Site-to-Site VPN

- Internally, the AWS Network Firewall uses the AWS Gateway Load Balancer
- Rules can be centrally managed cross-account by AWS Firewall Manager to apply to many VPCs



Network Firewall – Fine Grained Controls



- Supports 1000s of rules
  - IP & port - example: 10,000s of IPs filtering
  - Protocol – example: block the SMB protocol for outbound communications
  - Stateful domain list rule groups: only allow outbound traffic to *.mycorp.com or third-party software repo
  - General pattern matching using regex
- **Traffic filtering: Allow, drop, or alert for the traffic that matches the rules**
- **Active flow inspection** to protect against network threats with intrusion-prevention capabilities (like Gateway Load Balancer, but all managed by AWS)

- Send logs of rule matches to Amazon S3, CloudWatch Logs, Kinesis Data Firehose